

Lecture Notes: MATH 3603–Probability Theory

Zhigang Bao

Course Information

Textbook S. M. Ross, *Introduction to Probability Models*, 9th edition, Chapters 1–5, together with the lecture notes

Office hour Wednesday, 2:30–5:30 p.m.

Assessment component	Scheme 1 weight	Scheme 2 weight
Final examination	50%	90%
Two quizzes	40%	0%
Assignments	10%	10 %

Final grade= $\max\{\text{Scheme 1 grade}, \text{Scheme 2 grade}\}$

Contents

Course Information	1
 Chapter 1: Foundations of Probability	 6
1.1 Sample Spaces, Events, and Probability	6
1.1.1 Experiments, trials, and outcomes	6
1.1.2 Sample spaces	6
1.1.3 Events	7
1.1.4 Set operations on events	8
1.1.5 Probability and probability spaces	8
1.1.6 Basic consequences of the probability axioms	9
1.1.7 The inclusion–exclusion formula	9
 1.2 Conditional Probability	 11
1.2.1 Motivation: updating after new information	11
1.2.2 Conditional probability is itself a probability	11
1.2.3 The multiplication rule	12
1.2.4 Partitions and the law of total probability	12
1.2.5 Bayes’ theorem	12
1.2.6 Examples of conditional probability	13
1.2.7 Information may decrease or leave a probability unchanged	14
 1.3 Independence	 15
1.3.1 Independence of two events	15
1.3.2 Two common misconceptions	15
1.3.3 Independence of several events	16
1.3.4 Testing independence with dice	16
1.3.5 Product probability spaces	17
1.3.6 A dependent experiment: sampling without replacement	18
 Chapter 1 Summary of Key Formulas	 19
 Chapter 2: Random Variables, Distributions, and Limit Theorems	 21
2.1 Random Variables and Distribution Functions	21
2.1.1 Random variables as numerical descriptions of outcomes	21
2.1.2 Discrete and continuous behavior	22
2.1.3 Cumulative distribution functions	22
2.1.4 Probability mass functions	23
 2.2 Important Discrete Distributions	 24
2.2.1 Bernoulli trials and indicator variables	24
2.2.2 The binomial distribution	24
2.2.3 Geometric and negative binomial distributions	25
2.2.4 The Poisson distribution	25

2.2.5	Poisson approximation to the binomial law	25
2.3	Continuous Random Variables	26
2.3.1	Probability density functions	26
2.3.2	The uniform distribution	27
2.3.3	Inverse transform sampling	27
2.3.4	Exponential and gamma distributions	27
2.3.5	The Gaussian or normal distribution	28
2.4	Expectation, Moments, and Variance	29
2.4.1	Expectation as a theoretical average	29
2.4.2	Expectation of a function: LOTUS	30
2.4.3	Moments, variance, and standard deviation	30
2.5	Random Vectors and Joint Distributions	31
2.5.1	Random vectors	31
2.5.2	Joint cumulative distribution functions and marginals	32
2.5.3	Jointly discrete random variables	32
2.5.4	Jointly continuous random variables	33
2.5.5	Expectation of functions of random vectors	33
2.5.6	The coupon collector problem	34
2.6	Independence, Covariance, and Sums of Random Variables	34
2.6.1	Independence of random variables	34
2.6.2	Covariance and variance of a sum	35
2.6.3	Uncorrelated does not imply independent	35
2.6.4	Distribution of a sum and convolution	36
2.7	Moment Generating Functions	37
2.7.1	Definition and domain of existence	37
2.7.2	Independent sums	38
2.7.3	Moments from derivatives	38
2.7.4	Examples and applications	38
2.8	Limit Theorems	39
2.8.1	Markov's inequality	39
2.8.2	Chebyshev's inequality	40
2.8.3	Convergence in probability	40
2.8.4	Weak law of large numbers	40
2.8.5	Central limit theorem	41
Chapter 2 Summary of Key Formulas		42
Chapter 3: Conditional Distributions, Conditional Expectation, and Trans-		
formations		44
3.1	Conditional Distributions in the Discrete Case	44
3.1.1	From conditional probability to a conditional distribution	44

3.1.2	A Poisson conditioning identity	45
3.2	Conditional Expectation and the Law of Total Expectation	45
3.2.1	Conditioning on a value	45
3.2.2	Conditional expectation as a best prediction	46
3.2.3	The law of total expectation	47
3.2.4	Examples based on total expectation	47
3.3	Conditional Distributions and Expectations in the Continuous Case	48
3.3.1	The density-ratio formula	48
3.3.2	Conditional expectation in the continuous case	49
3.3.3	The bivariate normal distribution	49
3.4	An Optimal-Stopping Application: The Best-Prize Problem	50
3.5	Functions and Transformations of Random Variables	51
3.5.1	The preimage method	51
3.5.2	A strictly monotone transformation	52
3.5.3	Two-dimensional transformations and Jacobians	52
3.5.4	The distribution as a pushforward measure	53
	Chapter 3 Summary of Key Formulas	55
	Chapter 4: Discrete-Time Markov Chains	57
4.1	Stochastic Processes and the Markov Property	57
4.1.1	Stochastic processes	57
4.1.2	The Markov property	57
4.1.3	Basic examples	58
4.2	Transition Matrices and Multi-Step Transitions	59
4.2.1	The transition matrix	59
4.2.2	The Chapman–Kolmogorov equations	60
4.2.3	Evolution of an initial distribution	61
4.2.4	Two nearest-neighbor chains	61
4.3	Communicating Classes and Irreducibility	61
4.3.1	Accessibility and communication	61
4.4	Recurrence and Transience	62
4.4.1	First visits and first returns	63
4.4.2	The number of visits	63
4.4.3	Recurrence is a class property	64
4.5	Recurrence of Random Walks	64
4.5.1	One-dimensional nearest-neighbor random walk	64
4.5.2	Symmetric random walks in dimensions two and three	65
4.5.3	Finite irreducible chains	65

4.6	Stationary and Limiting Distributions	66
4.6.1	Stationarity	66
4.6.2	Two obstructions to convergence	66
4.6.3	Period and aperiodicity	67
4.6.4	Positive and null recurrence	67
4.6.5	The convergence theorem	68
4.6.6	Examples of stationary distributions	68
4.6.7	Long-run averages and rewards	69
4.7	Application: The Gambler’s Ruin Problem	69
4.7.1	The model and state classification	69
4.7.2	Probability of Reaching N before 0	70
4.7.3	Letting the upper boundary tend to infinity	71
4.8	Mean Time Spent in Transient States	71
4.8.1	The transient submatrix and the fundamental matrix	71
4.8.2	Example: occupation times in gambler’s ruin	72
4.8.3	Hitting probabilities from occupation times	73
4.9	The Galton–Watson Branching Process	73
4.9.1	Definition and elementary state classification	74
4.9.2	Mean and variance of the population	74
4.9.3	The extinction event	75
4.10	Probability Generating Functions	75
4.10.1	Definition and basic identities	75
4.10.2	Sums of independent random variables	76
4.10.3	A random number of summands	77
4.11	Extinction via Probability Generating Functions	78
4.12	Return-Time Generating Functions and Null Recurrence	80
	Chapter 4 Summary of Key Formulas	82
	Chapter 5: Poisson Processes	86
5.1	The Exponential Distribution and Memorylessness	86
5.1.1	Basic formulas	86
5.1.2	The memoryless property	86
5.1.3	An infinitesimal characterization	87
5.2	Counting Processes and the Poisson Process	88
5.2.1	Counting processes	88
5.2.2	First definition of a homogeneous Poisson process	88
5.2.3	Second definition: local behavior	89
5.3	Interarrival Times and Arrival Times	90
5.3.1	Interarrival times	90

5.3.2	The Waiting Time Until the n th Event	91
5.4	Splitting, Thinning, and Superposition	92
5.4.1	Independent classification of events	92
5.4.2	Superposition	93
5.5	Applications of Splitting and Exponential Waiting Times	93
5.5.1	Poissonization of the coupon collector problem	93
5.5.2	Types appearing exactly once	94
5.5.3	Races between independent exponential clocks	95
5.5.4	Which Poisson process reaches its target first?	96
5.6	Conditional Arrival Times and Order Statistics	96
5.6.1	One arrival conditioned on the count	96
5.6.2	Order statistics	96
5.6.3	Arrival Times Conditioned on the Count	97
5.6.4	Earlier Arrivals Conditioned on the n th Arrival Time	98
5.7	Nonhomogeneous Poisson Processes	98
5.7.1	Time-dependent intensity	98
5.7.2	Time-dependent thinning	99
5.7.3	Superposition and identification of the source	99
5.7.4	Time-dependent classification into several types	100
5.7.5	The mean-value function	100
Chapter 5 Summary of Key Formulas		102

Chapter 1

Foundations of Probability

Scope of Chapter 1

This chapter introduces the mathematical language of probability: experiments, sample spaces, events, set operations, and the axioms of probability. It develops the principal consequences of those axioms, including inclusion–exclusion, before turning to conditional probability, the multiplication rule, the law of total probability, and Bayes’ theorem. The chapter concludes with independence, the distinction between pairwise and mutual independence, product probability spaces, and sampling with and without replacement.

1.1 Sample Spaces, Events, and Probability

1.1.1 Experiments, trials, and outcomes

Probability begins with an *experiment* (or *trial*) whose result is not known in advance. Each possible result is called an *outcome*.

Experiment	Possible outcomes	Typical probabilities
Flip a fair coin	H or T	$\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}$
Roll a fair die	1, 2, 3, 4, 5, 6	Each face has probability $\frac{1}{6}$
Observe the lifetime of a bulb	A value $w \in [0, \infty)$	(to be specified in a later stage)

The first two experiments have finitely many possible outcomes. The bulb-lifetime experiment has continuously many possible outcomes.

1.1.2 Sample spaces

Definition 1.1.1 (Sample space). The *sample space* of an experiment is the set of all possible outcomes. It is usually denoted by S or Ω .

Example 1.1.2 (One coin). For a single coin flip,

$$S = \{H, T\}.$$

Example 1.1.3 (Two coins). For two successive coin flips, it is convenient to record the order:

$$S = \{(H, H), (H, T), (T, H), (T, T)\}.$$

The same sample space may be written more briefly as

$$S = \{HH, HT, TH, TT\}.$$

Example 1.1.4 (Lifetime of a bulb). If the lifetime is measured in a nonnegative unit of time, then

$$S = [0, \infty).$$

1.1.3 Events

A single outcome is not always the object of interest. Often we ask whether the outcome belongs to a specified collection of outcomes.

Definition 1.1.5 (Event). In the setting of these notes, an *event* is any subset of the sample space S .

An event containing exactly one outcome is called an *elementary event* or a *simple event*. An event containing several outcomes is called a *compound event*.

Example 1.1.6 (Events for a die roll). Let

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Then:

- $\{4\}$ is an elementary event;
- $\{2, 4, 6\}$ is the event that the die shows an even number;
- $\{4, 5, 6\}$ is the event that the die shows a relatively large number.

Example 1.1.7 (Events for two coin flips). For

$$S = \{HH, HT, TH, TT\},$$

the event $\{HH\}$ is elementary, while

$$E = \{HH, HT\}$$

is the compound event that the first coin is heads.

Event language

An event E *occurs* precisely when the observed outcome ω belongs to E :

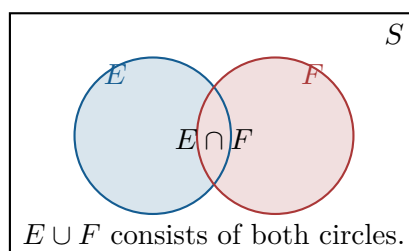
$$E \text{ occurs} \iff \omega \in E.$$

Thus, “the probability of the event E ,” “the probability that E occurs,” and “the probability that the outcome lies in E ” are equivalent expressions.

1.1.4 Set operations on events

Because events are sets, ordinary set operations describe combinations of events.

Notation	Meaning
$E \cup F$	At least one of E and F occurs; equivalently, either E or F occurs, possibly both.
$E \cap F$	Both E and F occur.
$E^c = S \setminus E$	E does not occur.
$E \cap F = \emptyset$	E and F are disjoint, or mutually exclusive; they cannot occur together.



1.1.5 Probability and probability spaces

Probability assigns a numerical measure of likelihood to events. Thus probability is a function whose inputs are events.

Definition 1.1.8 (Probability). Let S be a sample space and let $\mathcal{F}$ be the class of events under consideration. A probability is a set function

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$$

satisfying the following conditions:

- (i) $0 \leq \mathbb{P}(A) \leq 1$ for every event A ;
- (ii) $\mathbb{P}(S) = 1$;
- (iii) if $A_1, A_2, \dots$ are mutually exclusive events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The third condition is called *countable additivity*.

Definition 1.1.9 (Probability space). A probability space is written as

$$(S, \mathcal{F}, \mathbb{P}),$$

where S is the sample space, $\mathcal{F}$ is the class of events, and $\mathbb{P}$ is the probability function.

The notation $\mathbb{P}(E)$ is meaningful because E is an event. Writing $\mathbb{P}(2)$ without first interpreting 2 as an event is not meaningful. For a die roll, for example, one should write $\mathbb{P}(\{2\})$.

1.1.6 Basic consequences of the probability axioms

Proposition 1.1.10. *For events $A, B, A_1, \dots, A_n$, the following properties hold.*

(1) $\mathbb{P}(\emptyset) = 0$.

(2) If $A_1, \dots, A_n$ are mutually exclusive, then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i).$$

(3) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

(4) If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof. For (1), apply countable additivity to a sequence consisting entirely of the empty set. Since the union is again $\emptyset$,

$$\mathbb{P}(\emptyset) = \sum_{i=1}^{\infty} \mathbb{P}(\emptyset),$$

which forces $\mathbb{P}(\emptyset) = 0$.

For (2), set $A_{n+1} = A_{n+2} = \dots = \emptyset$ and apply countable additivity.

For (3), the events A and A^c are mutually exclusive and satisfy $A \cup A^c = S$. Hence

$$1 = \mathbb{P}(S) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

For (4), if $A \subseteq B$, then

$$B = A \cup (B \setminus A),$$

and the two sets on the right are disjoint. Here $B \setminus A = B \cap (A^c)$. Therefore

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A).$$

□

1.1.7 The inclusion–exclusion formula

Theorem 1.1.11 (Two-event inclusion–exclusion). *For any two events A and B ,*

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).}$$

Proof. Decompose the union into three mutually exclusive pieces:

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

Also,

$$A = (A \setminus B) \cup (A \cap B), \quad B = (B \setminus A) \cup (A \cap B).$$

Adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts $A \cap B$ twice, so one copy must be subtracted. □

If $A \cap B = \emptyset$, the intersection term vanishes and the formula reduces to finite additivity.

Theorem 1.1.12 (General inclusion–exclusion). *For events $A_1, \dots, A_n$,*

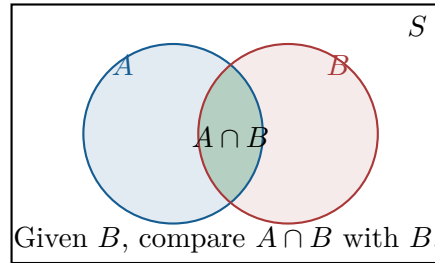
$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) \\ &\quad + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots \\ &\quad + (-1)^{n+1} \mathbb{P}(A_1 \cap \dots \cap A_n).\end{aligned}$$

The general formula may be proved by induction on n .

1.2 Conditional Probability

1.2.1 Motivation: updating after new information

Suppose we are interested in an event A , but we are then told that another event B has occurred. The information B restricts the possible outcome to the reduced sample space B . Within this reduced sample space, A occurs precisely through the portion $A \cap B$.



Definition 1.2.1 (Conditional probability). If $\mathbb{P}(B) > 0$, the conditional probability that A occurs given that B has occurred is

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This is read as “the probability of A given B ” or “the probability of A conditioned on B .”

1.2.2 Conditional probability is itself a probability

Proposition 1.2.2. Fix an event B with $\mathbb{P}(B) > 0$. The set function

$$Q_B(A) := \mathbb{P}(A \mid B)$$

is a probability function on the events A .

Proof. First,

$$0 \leq \mathbb{P}(A \cap B) \leq \mathbb{P}(B),$$

so $0 \leq Q_B(A) \leq 1$. Next,

$$Q_B(S) = \frac{\mathbb{P}(S \cap B)}{\mathbb{P}(B)} = 1.$$

Finally, if $A_1, A_2, \dots$ are mutually exclusive, then $A_1 \cap B, A_2 \cap B, \dots$ are also mutually exclusive and

$$\left(\bigcup_{i=1}^{\infty} A_i \right) \cap B = \bigcup_{i=1}^{\infty} (A_i \cap B).$$

Therefore

$$\begin{aligned} Q_B\left(\bigcup_{i=1}^{\infty} A_i\right) &= \frac{\mathbb{P}\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{\mathbb{P}(B)} \\ &= \sum_{i=1}^{\infty} \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_{i=1}^{\infty} Q_B(A_i). \end{aligned}$$

□

1.2.3 The multiplication rule

Rearranging the definition gives the useful identity

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B)\mathbb{P}(B).$$

Similarly, when $\mathbb{P}(A) > 0$,

$$\mathbb{P}(A \cap B) = \mathbb{P}(B | A)\mathbb{P}(A).$$

1.2.4 Partitions and the law of total probability

Definition 1.2.3 (Partition). Events $F_1, \dots, F_n$ form a partition of S if they are mutually exclusive and

$$\bigcup_{i=1}^n F_i = S.$$

Theorem 1.2.4 (Law of total probability). Suppose $F_1, \dots, F_n$ form a partition of S and $\mathbb{P}(F_i) > 0$ for every i . Then, for any event E ,

$$\mathbb{P}(E) = \sum_{j=1}^n \mathbb{P}(F_j)\mathbb{P}(E | F_j).$$

Proof. Because the F_j form a partition,

$$E = E \cap S = \bigcup_{j=1}^n (E \cap F_j),$$

and the events $E \cap F_j$ are mutually exclusive. Hence

$$\mathbb{P}(E) = \sum_{j=1}^n \mathbb{P}(E \cap F_j) = \sum_{j=1}^n \mathbb{P}(F_j)\mathbb{P}(E | F_j).$$

□

1.2.5 Bayes' theorem

Theorem 1.2.5 (Bayes' theorem). Suppose $F_1, \dots, F_n$ form a partition of S , with $\mathbb{P}(F_j) > 0$ for every j . If $\mathbb{P}(E) > 0$, then

$$\mathbb{P}(F_i | E) = \frac{\mathbb{P}(F_i)\mathbb{P}(E | F_i)}{\sum_{j=1}^n \mathbb{P}(F_j)\mathbb{P}(E | F_j)}.$$

Proof. By conditional probability and the multiplication rule,

$$\mathbb{P}(F_i | E) = \frac{\mathbb{P}(F_i \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(F_i)\mathbb{P}(E | F_i)}{\mathbb{P}(E)}.$$

Apply the law of total probability to the denominator. □

Why Bayes' theorem is useful

In many applications, the prior probabilities $\mathbb{P}(F_j)$ and the likelihoods $\mathbb{P}(E \mid F_j)$ are known, while the desired quantity is the reverse conditional probability $\mathbb{P}(F_i \mid E)$.

1.2.6 Examples of conditional probability

Example 1.2.6 (A family with two children). Assume the ordered possibilities are equally likely:

$$S = \{(B, B), (B, G), (G, B), (G, G)\},$$

where the first and second coordinates distinguish the two children. Let

$$E = \{(B, B)\}$$

be the event that both children are boys, and let

$$F = \{(B, B), (B, G), (G, B)\}$$

be the event that at least one child is a boy. Since $E \subseteq F$,

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E)}{\mathbb{P}(F)} = \frac{1/4}{3/4} = \boxed{\frac{1}{3}}.$$

Example 1.2.7 (Choosing between two urns). Urn 1 contains 2 white balls and 7 black balls. Urn 2 contains 5 white balls and 6 black balls. A fair coin is flipped:

- if the result is H , one ball is selected from urn 1;
- if the result is T , one ball is selected from urn 2.

Let W be the event that a white ball is selected. We seek $\mathbb{P}(H \mid W)$.

The relevant probabilities are

$$\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}, \quad \mathbb{P}(W \mid H) = \frac{2}{9}, \quad \mathbb{P}(W \mid T) = \frac{5}{11}.$$

Bayes' theorem gives

$$\begin{aligned} \mathbb{P}(H \mid W) &= \frac{\mathbb{P}(H)\mathbb{P}(W \mid H)}{\mathbb{P}(H)\mathbb{P}(W \mid H) + \mathbb{P}(T)\mathbb{P}(W \mid T)} \\ &= \frac{\frac{1}{2} \cdot \frac{2}{9}}{\frac{1}{2} \cdot \frac{2}{9} + \frac{1}{2} \cdot \frac{5}{11}} = \boxed{\frac{22}{67}}. \end{aligned}$$

The combined experiment has four qualitative outcomes,

$$S = \{(H, W), (H, B), (T, W), (T, B)\},$$

but these outcomes are not equally likely. Their probabilities are

$$\mathbb{P}(H, W) = \frac{1}{9}, \quad \mathbb{P}(H, B) = \frac{7}{18}, \quad \mathbb{P}(T, W) = \frac{5}{22}, \quad \mathbb{P}(T, B) = \frac{3}{11}.$$

1.2.7 Information may decrease or leave a probability unchanged

Additional information does not necessarily increase the probability of the event under study.

Example 1.2.8 (Information that lowers the probability). A multiple-choice question has choices a, b, c, d , and a is the correct answer. If one chooses uniformly at random, then

$$\mathbb{P}(\text{correct}) = \mathbb{P}(\{a\}) = \frac{1}{4}.$$

After learning that the selected answer lies in $\{b, c, d\}$,

$$\mathbb{P}(\{a\} \mid \{b, c, d\}) = 0.$$

Example 1.2.9 (Information that has no effect). Flip two fair coins. Let

$$E = \{HH, HT\}$$

be the event that the first coin is heads, and let

$$F = \{HT, TT\}$$

be the event that the second coin is tails. Then

$$\mathbb{P}(E) = \frac{1}{2}, \quad \mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{1/4}{1/2} = \frac{1}{2}.$$

Thus learning F does not change the probability of E . This leads to the notion of independence.

1.3 Independence

1.3.1 Independence of two events

Definition 1.3.1 (Independent events). Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

We may write $A \perp\!\!\!\perp B$. If A and B are not independent, we say that they are dependent.

Proposition 1.3.2. *If $\mathbb{P}(B) > 0$, then A and B are independent if and only if*

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

If $\mathbb{P}(A) > 0$, independence is also equivalent to

$$\mathbb{P}(B \mid A) = \mathbb{P}(B).$$

Proof. When $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Thus $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ exactly when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

The second equivalence is identical with A and B interchanged. □

1.3.2 Two common misconceptions

Independence is not mutual exclusivity

The notation $A \perp\!\!\!\perp B$ does *not* mean $A \cap B = \emptyset$. In fact, if A and B are mutually exclusive and both have positive probability, then

$$\mathbb{P}(A \cap B) = 0 \quad \text{but} \quad \mathbb{P}(A)\mathbb{P}(B) > 0,$$

so they are not independent.

Independence is not an informal judgment

It is not enough to say that two events seem “unrelated,” nor does apparent “relatedness” prove dependence. Independence is determined by the exact identity

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

1.3.3 Independence of several events

Definition 1.3.3 (Mutual or joint independence). Events $A_1, \dots, A_n$ are mutually independent if, for every subset

$$J \subseteq \{1, \dots, n\}, \quad |J| \geq 2,$$

one has

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j).$$

Checking only pairs gives *pairwise independence*, which is weaker than mutual independence.

Example 1.3.4 (Pairwise independent but not mutually independent). Two fair dice are rolled. Define

$$\begin{aligned} A &= \{\text{the sum is 7}\}, \\ B &= \{\text{the first die is 4}\}, \\ C &= \{\text{the second die is 3}\}. \end{aligned}$$

Each event has probability $1/6$. Moreover,

$$A \cap B = A \cap C = B \cap C = \{(4, 3)\},$$

so every pair has intersection probability

$$\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}.$$

Thus the events are pairwise independent. However,

$$A \cap B \cap C = \{(4, 3)\},$$

and therefore

$$\mathbb{P}(A \cap B \cap C) = \frac{1}{36} \neq \frac{1}{216} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

Hence they are not mutually independent.

1.3.4 Testing independence with dice

Roll two fair dice, so the sample space consists of the 36 ordered pairs (i, j) , each with probability $1/36$.

Example 1.3.5 (The sum is 6 and the first die is 4). Let

$$\begin{aligned} A_6 &= \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}, \\ B &= \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\}. \end{aligned}$$

Then

$$A_6 \cap B = \{(4, 2)\},$$

so

$$\mathbb{P}(A_6 \cap B) = \frac{1}{36}, \quad \mathbb{P}(A_6)\mathbb{P}(B) = \frac{5}{36} \cdot \frac{1}{6} = \frac{5}{216}.$$

These quantities are different, so A_6 and B are not independent.

Example 1.3.6 (The sum is 7 and the first die is 4). Let

$$A_7 = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Then $A_7 \cap B = \{(4, 3)\}$ and

$$\mathbb{P}(A_7 \cap B) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = \mathbb{P}(A_7)\mathbb{P}(B).$$

Therefore A_7 and B are independent.

1.3.5 Product probability spaces

Suppose two experiments are performed independently. Let their probability spaces be

$$(S_1, \mathcal{F}_1, \mathbb{P}_1) \quad \text{and} \quad (S_2, \mathcal{F}_2, \mathbb{P}_2).$$

The combined experiment has sample space

$$S = S_1 \times S_2 = \{(\omega_1, \omega_2) : \omega_1 \in S_1, \omega_2 \in S_2\}.$$

For rectangular events $E_1 \times E_2$, the product probability is

$$\boxed{\mathbb{P}(E_1 \times E_2) = \mathbb{P}_1(E_1)\mathbb{P}_2(E_2).}$$

Example 1.3.7 (A coin flip and a die roll). Let

$$S_1 = \{H, T\}, \quad S_2 = \{1, 2, 3, 4, 5, 6\}.$$

Take

$$E_1 = \{H\}, \quad E_2 = \{2, 4, 6\}.$$

The event that the coin is heads, with the die arbitrary, is

$$A = E_1 \times S_2.$$

The event that the die is even, with the coin arbitrary, is

$$B = S_1 \times E_2.$$

Their intersection is

$$A \cap B = E_1 \times E_2.$$

Consequently,

$$\mathbb{P}(A \cap B) = \mathbb{P}_1(E_1)\mathbb{P}_2(E_2) = \mathbb{P}(A)\mathbb{P}(B),$$

so A and B are independent.

1.3.6 A dependent experiment: sampling without replacement

Consider a bag containing one red ball and three blue balls. Draw one ball and, without replacing it, draw a second ball. Although each draw has the color labels

$$S_1 = S_2 = \{R, B\},$$

the pair (R, R) is impossible because the bag contains only one red ball. Thus the actual color-pair support is not the full Cartesian product $S_1 \times S_2$.

For example,

$$\mathbb{P}(\text{second draw is red} \mid \text{first draw is red}) = 0,$$

whereas the unconditional probability that the second draw is red is $1/4$. Hence the two draws are not independent.

Chapter 1 Summary of Key Formulas

Probability axioms and basic rules

For any event A ,

$$0 \leq \mathbb{P}(A) \leq 1, \quad \mathbb{P}(S) = 1, \quad \mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

If $A \subseteq B$, then

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

For pairwise disjoint events $A_1, A_2, \dots$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Inclusion–exclusion

For two events,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

More generally,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) + \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} \mathbb{P}\left(\bigcap_{i=1}^n A_i\right).$$

Conditional probability, total probability, and Bayes' theorem

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

If $F_1, \dots, F_n$ form a partition of S , then

$$\mathbb{P}(E) = \sum_{j=1}^n \mathbb{P}(F_j)\mathbb{P}(E \mid F_j).$$

When $\mathbb{P}(E) > 0$,

$$\mathbb{P}(F_i \mid E) = \frac{\mathbb{P}(F_i)\mathbb{P}(E \mid F_i)}{\sum_{j=1}^n \mathbb{P}(F_j)\mathbb{P}(E \mid F_j)}.$$

Independence

Events A and B are independent precisely when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

When the relevant conditional probabilities are defined, this is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \quad \text{and} \quad \mathbb{P}(B \mid A) = \mathbb{P}(B).$$

For mutual independence of $A_1, \dots, A_n$, the product identity must hold for every subcollection of at least two events:

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j), \quad \emptyset \neq J \subseteq \{1, \dots, n\}, \quad |J| \geq 2.$$

For independent experiments on product spaces,

$$\mathbb{P}(E_1 \times E_2) = \mathbb{P}_1(E_1)\mathbb{P}_2(E_2).$$

Chapter 2

Random Variables, Distributions, and Limit Theorems

Scope of Chapter 2

This chapter develops random variables and their distributions through cumulative distribution functions, probability mass functions, and probability density functions. It presents the main discrete and continuous families used in the course, then studies expectation, moments, variance, random vectors, joint and marginal distributions, independence, covariance, and sums of random variables. Moment-generating functions provide a systematic tool for identifying distributions, and the chapter concludes with the weak law of large numbers and the central limit theorem.

2.1 Random Variables and Distribution Functions

2.1.1 Random variables as numerical descriptions of outcomes

A random experiment produces an outcome $\omega \in S$. A random variable assigns a numerical value to that outcome.

Definition 2.1.1 (Random variable). Let $(S, \mathcal{F}, \mathbb{P})$ be a probability space. In this lecture notes, a *real-valued random variable* is simply defined as a function

$$X : S \longrightarrow \mathbb{R}.$$

Remark 2.1.2. In measure-theoretic probability, we need additional restriction on the function X to regard it as a random variable.

Because the outcome ω is uncertain, the value $X(\omega)$ is uncertain as well.

Example 2.1.3 (Number of heads in two tosses). For two fair coin tosses,

$$S = \{HH, HT, TH, TT\}.$$

Define X to be the number of heads. Then

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

Thus X takes values in $\{0, 1, 2\}$.

Example 2.1.4 (Cutting a rope). A rope of length 1 is cut at a uniformly chosen point $\omega \in [0, 1]$. Two natural random variables are

$$X(\omega) = \omega, \quad L(\omega) = \min\{\omega, 1 - \omega\},$$

where X records the cut location and L records the length of the shorter piece.

For any set $B \subseteq \mathbb{R}$, the notation

$$\{X \in B\} := \{\omega \in S : X(\omega) \in B\}$$

denotes an event in the original sample space. In particular,

$$\mathbb{P}(X = x) = \mathbb{P}(\{\omega : X(\omega) = x\}).$$

2.1.2 Discrete and continuous behavior

Example 2.1.5 (Two fair dice). Roll two fair dice and let X be their sum. Then $X \in \{2, 3, \dots, 12\}$ and

$$\mathbb{P}(X = 2) = \mathbb{P}(\{(1, 1)\}) = \frac{1}{36}.$$

Let Y be the number of dice showing 3. Then $Y \in \{0, 1, 2\}$, and

$$\mathbb{P}(Y = 2) = \frac{1}{36}, \quad \mathbb{P}(Y = 0) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}.$$

For a continuously distributed random variable, a single point usually has probability zero. For example, if the cut location X is uniform on $[0, 1]$, then

$$\mathbb{P}(X = x) = 0 \quad \text{for every fixed } x \in [0, 1].$$

However, this single point probability does not tell us more information about how X is distributed on the real line. Consequently, the informative questions concern intervals or more general sets, such as

$$\mathbb{P}(a < X \leq b), \quad \mathbb{P}(X \in [a, b]), \quad \mathbb{P}(X \leq b).$$

2.1.3 Cumulative distribution functions

Definition 2.1.6 (Cumulative distribution function). The *cumulative distribution function*, or *c.d.f.*, of a random variable X is

$$F_X(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

The c.d.f. records the probability of every left half-line. It therefore determines all interval probabilities. For example, if $a < b$, then

$$\begin{aligned} \mathbb{P}(a < X \leq b) &= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) \\ &= F_X(b) - F_X(a). \end{aligned}$$

Proposition 2.1.7 (Basic properties of a c.d.f.). *Every c.d.f. F_X has the following properties.*

- (i) F_X is nondecreasing.
- (ii) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$.
- (iii) F_X is right-continuous:

$$F_X(x) = \lim_{h \downarrow 0} F_X(x + h).$$

(iv) *The left limit*

$$F_X(x^-) := \lim_{h \downarrow 0} F_X(x - h) = \mathbb{P}(X < x)$$

exists for every x .

(v) *The jump of the c.d.f. at x is the point mass at x :*

$$\mathbb{P}(X = x) = F_X(x) - F_X(x^-).$$

Proof. If $a \leq b$, then $\{X \leq a\} \subseteq \{X \leq b\}$, which gives monotonicity. The limits at $\pm\infty$ and the one-sided continuity statements follow from continuity of probability for increasing and decreasing sequences of events. Finally,

$$\{X \leq x\} = \{X < x\} \cup \{X = x\},$$

so subtracting probabilities gives the jump formula. □

A useful consequence is

$$\boxed{\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a-).}$$

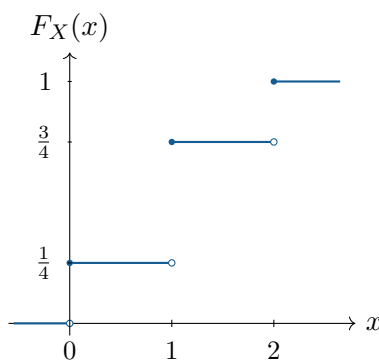
Example 2.1.8 (C.d.f. for the number of heads). For the number X of heads in two fair tosses,

$$\mathbb{P}(X = 0) = \frac{1}{4}, \quad \mathbb{P}(X = 1) = \frac{1}{2}, \quad \mathbb{P}(X = 2) = \frac{1}{4}.$$

Hence

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \leq x < 1, \\ \frac{3}{4}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

The c.d.f. is a right-continuous step function whose jumps are precisely the point probabilities.



2.1.4 Probability mass functions

Definition 2.1.9 (Discrete random variable and p.m.f.). A random variable X is *discrete* if it takes values in a finite or countably infinite set. Its *probability mass function*, or *p.m.f.*, is

$$p_X(x) := \mathbb{P}(X = x).$$

If the possible values are $x_1, x_2, \dots$, then

$$p_X(x_i) \geq 0, \quad \sum_i p_X(x_i) = 1, \quad F_X(b) = \sum_{x_i \leq b} p_X(x_i).$$

Conversely,

$$p_X(x) = F_X(x) - F_X(x^-).$$

2.2 Important Discrete Distributions

2.2.1 Bernoulli trials and indicator variables

A *Bernoulli trial* is an experiment with two designated outcomes: success and failure. Let

$$\mathbb{P}(\text{success}) = p, \quad \mathbb{P}(\text{failure}) = 1 - p.$$

Definition 2.2.1 (Bernoulli distribution). A random variable X has the Bernoulli distribution with parameter p , written

$$X \sim \text{Bernoulli}(p),$$

if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

Here X records whether a single Bernoulli trial is successful.

Definition 2.2.2 (Indicator random variable). For an event A , its indicator is

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Then $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$.

2.2.2 The binomial distribution

Definition 2.2.3 (Binomial distribution). Suppose n independent Bernoulli trials are performed, each having success probability p . Let X be the number of successes. Then

$$X \sim \text{Bin}(n, p)$$

and

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Indeed, one chooses the k successful trials in $\binom{n}{k}$ ways, and each specified success–failure pattern has probability $p^k (1 - p)^{n-k}$.

2.2.3 Geometric and negative binomial distributions

Definition 2.2.4 (Geometric distribution). Let X be the number of Bernoulli trials required to obtain the first success. Then

$$X \sim \text{Geom}(p), \quad \mathbb{P}(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

Definition 2.2.5 (Negative binomial distribution). Let X be the number of Bernoulli trials required to obtain the r th success. Then

$$X \sim \text{NegBin}(r, p), \quad X \in \{r, r + 1, \dots\},$$

and

$$\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r + 1, \dots$$

The factor $\binom{k-1}{r-1}$ chooses the locations of the first $r - 1$ successes among the first $k - 1$ trials; the k th trial must be the r th success.

2.2.4 The Poisson distribution

Definition 2.2.6 (Poisson distribution). For $\lambda > 0$, a random variable X has the Poisson distribution with parameter λ , written

$$X \sim \text{Poisson}(\lambda),$$

if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Poisson random variables are commonly used to model counts of relatively rare events during a fixed time interval or in a fixed region, such as the number of misprints on a page or the number of accidents during a specified period.

2.2.5 Poisson approximation to the binomial law

The Poisson law arises as a limit of binomial laws when the number of trials is large, the individual success probability is small, and their product remains moderate.

Proposition 2.2.7 (Poisson limit). Let $X_n \sim \text{Bin}(n, p_n)$ with

$$np_n \longrightarrow \lambda \in (0, \infty).$$

Then, for every fixed $k \geq 0$,

$$\mathbb{P}(X_n = k) \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

In particular, X_n converges in distribution to $\text{Poisson}(\lambda)$.

Proof. It is enough to consider $p_n = \lambda/n$; the general case is analogous. For fixed k ,

$$\begin{aligned}\mathbb{P}(X_n = k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{n(n-1)\cdots(n-k+1)}{n^k} \frac{\lambda^k}{k!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}.\end{aligned}$$

The first and last factors tend to 1, while

$$\left(1 - \frac{\lambda}{n}\right)^n \longrightarrow e^{-\lambda}.$$

□

2.3 Continuous Random Variables

2.3.1 Probability density functions

Definition 2.3.1 (Continuous random variable and p.d.f.). A random variable X is called *continuous* in these notes if there is a nonnegative integrable function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(u) \, du.$$

The function f_X is called the *probability density function*, or *p.d.f.*, of X .

A density is not itself a probability. Rather, probability is obtained by integration:

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(u) \, du.$$

When f_X is continuous at x , the small-interval approximation is

$$\mathbb{P}(x < X \leq x + \Delta x) = \int_x^{x+\Delta x} f_X(u) \, du \approx f_X(x) \Delta x.$$

Proposition 2.3.2 (Basic properties of a density). *If X has p.d.f. f_X , then*

- (i) $f_X(x) \geq 0$ and $\int_{-\infty}^{\infty} f_X(x) \, dx = 1$;
- (ii) $\mathbb{P}(X = x) = 0$ for every fixed x ;
- (iii) for every interval $[a, b]$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx;$$

- (iv) more generally, for any sufficiently regular set $B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in B) = \int_B f_X(x) \, dx.$$

Whenever F_X is differentiable at x , one has $F_X'(x) = f_X(x)$.

2.3.2 The uniform distribution

Definition 2.3.3 (Uniform distribution). A random variable X is uniform on $[a, b]$, written

$$X \sim \text{Unif}[a, b], \quad a < b,$$

if

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Its c.d.f. is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

2.3.3 Inverse transform sampling

A c.d.f. can be used to generate a random variable with a prescribed distribution.

Proposition 2.3.4 (Inverse transform in the continuous invertible case). *Let $U \sim \text{Unif}[0, 1]$. Suppose F is a continuous, strictly increasing c.d.f., and define*

$$X = F^{-1}(U).$$

Then X has c.d.f. F .

Proof. For every x ,

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x),$$

because $\mathbb{P}(U \leq t) = t$ for $0 \leq t \leq 1$. □

2.3.4 Exponential and gamma distributions

Definition 2.3.5 (Exponential distribution). For $\lambda > 0$, a random variable X is exponential with rate λ , written

$$X \sim \text{Exp}(\lambda),$$

if

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Its c.d.f. is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

The exponential distribution models a waiting time for the first event in a Poisson process, in the same way that the geometric distribution models the number of Bernoulli trials required for the first success.

Definition 2.3.6 (Gamma distribution). For shape $\alpha > 0$ and rate $\lambda > 0$, a random variable X has the gamma distribution, written

$$X \sim \text{Gamma}(\alpha, \lambda),$$

if

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0$$

where

$$\Gamma(\alpha) = \int_0^\infty e^{-u} u^{\alpha-1} \, du.$$

When α is a positive integer, the gamma law describes the waiting time for the α th event in a Poisson process.

2.3.5 The Gaussian or normal distribution

Definition 2.3.7 (Normal distribution). A random variable X is normal with mean μ and variance parameter $\sigma^2 > 0$, written

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

For the standard normal law $Z \sim \mathcal{N}(0, 1)$, write

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(u) \, du.$$

The general normal density is obtained by shifting and rescaling:

$$f_X(x) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right).$$

Proposition 2.3.8 (The Gaussian density is normalized).

$$\int_{-\infty}^{\infty} \varphi(x) \, dx = 1.$$

Proof. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} \, dx.$$

Then

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)/2} \, dx \, dy.$$

Changing to polar coordinates gives

$$I^2 = \int_0^{2\pi} \int_0^\infty e^{-r^2/2} r \, dr \, d\theta = 2\pi.$$

Thus $I = \sqrt{2\pi}$, proving the claim. □

Proposition 2.3.9 (Affine transformations of a normal variable). *If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$ with $a \neq 0$, then*

$$Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

In particular, if $Z \sim \mathcal{N}(0, 1)$, then

$$\mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2).$$

Proof. For $a > 0$,

$$F_Y(y) = \mathbb{P}(aX + b \leq y) = F_X\left(\frac{y - b}{a}\right).$$

Differentiating gives

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y - b}{a}\right),$$

which is the stated normal density. The case $a < 0$ is handled similarly, with $|a|$ replacing a in the density transformation. $\square$

2.4 Expectation, Moments, and Variance

2.4.1 Expectation as a theoretical average

Suppose $X_1, \dots, X_N$ are numerical outcomes from repeated versions of the same experiment. The empirical average is

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i.$$

If a discrete value x occurs with frequency close to $Np_X(x)$, then formally

$$\bar{X}_N \approx \sum_x x p_X(x).$$

This motivates the definition of expectation.

Definition 2.4.1 (Expectation). If X is discrete with p.m.f. p_X , then

$$\mathbb{E}X = \sum_x x p_X(x),$$

whenever the sum is well-defined. If X is continuous with p.d.f. f_X , then

$$\mathbb{E}X = \int_{-\infty}^{\infty} x f_X(x) \, dx,$$

whenever the integral is well-defined.

Example 2.4.2 (Bernoulli mean). If $X \sim \text{Bernoulli}(p)$, then

$$\mathbb{E}X = 1 \cdot p + 0 \cdot (1 - p) = p.$$

Example 2.4.3 (Binomial mean by direct summation). If $X \sim \text{Bin}(n, p)$, then

$$\begin{aligned}\mathbb{E}X &= \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} \\ &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1-p)^{(n-1)-(k-1)} \\ &= np.\end{aligned}$$

Example 2.4.4 (Mean of a normal variable). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then symmetry after the substitution $y = x - \mu$ gives

$$\mathbb{E}X = \mu.$$

2.4.2 Expectation of a function: LOTUS

Let $g : \mathbb{R} \rightarrow \mathbb{R}$. To compute $\mathbb{E}[g(X)]$, it is unnecessary to first determine the distribution of $g(X)$.

Proposition 2.4.5 (Law of the unconscious statistician). *If X is discrete, then*

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

If X is continuous, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx.$$

Proof in the discrete case. Let $Y = g(X)$. Grouping together all values x for which $g(x) = y$ gives

$$\begin{aligned}\mathbb{E}Y &= \sum_y y \mathbb{P}(Y = y) \\ &= \sum_y y \sum_{x:g(x)=y} p_X(x) \\ &= \sum_x g(x) p_X(x).\end{aligned}$$

□

2.4.3 Moments, variance, and standard deviation

Definition 2.4.6 (Moments). The k th moment of X is

$$\mathbb{E}[X^k].$$

The k th central moment is

$$\mathbb{E}[(X - \mathbb{E}X)^k].$$

Definition 2.4.7 (Variance and standard deviation). The variance of X is

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2],$$

and the standard deviation is

$$\text{SD}(X) := \sqrt{\text{Var}(X)}.$$

Variance measures the dispersion of X around its mean.

Proposition 2.4.8 (Computational formula for variance). *Whenever the second moment is finite,*

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.}$$

Proof. Expanding the square,

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2] \\ &= \mathbb{E}[X^2] - 2(\mathbb{E}X)^2 + (\mathbb{E}X)^2.\end{aligned}$$

□

Proposition 2.4.9 (Variance under affine transformations). *For $a, b \in \mathbb{R}$,*

$$\boxed{\text{Var}(aX + b) = a^2 \text{Var}(X).}$$

Example 2.4.10 (Variance of a normal variable). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then after the substitutions $y = x - \mu$ and $t = y/\sigma$,

$$\text{Var}(X) = \sigma^2 \int_{-\infty}^{\infty} t^2 \varphi(t) \, dt = \sigma^2.$$

The final integral equals 1, for example by integration by parts.

Moments need not exist

A probability density may have heavy tails. For the standard Cauchy density

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R},$$

one has

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} \frac{x^2}{\pi(1+x^2)} \, dx = \infty.$$

Thus not every random variable has a finite mean or variance.

2.5 Random Vectors and Joint Distributions

2.5.1 Random vectors

Several numerical features of the same outcome can be studied simultaneously.

Definition 2.5.1 (Random vector). If $X_1, \dots, X_d$ are random variables on the same probability space, then

$$\mathbf{X} = (X_1, \dots, X_d) : S \rightarrow \mathbb{R}^d$$

is called a *random vector*. Its components are said to be jointly distributed.

Example 2.5.2 (Height and weight). Suppose one of five basketball candidates is chosen uniformly at random.

Candidate	Height H	Weight W
1	180	69
2	182	75
3	175	65
4	174	70
5	170	68

Then

$$\mathbb{P}(H \geq 175) = \frac{3}{5}, \quad \mathbb{P}(W \geq 70) = \frac{2}{5},$$

but only candidate 2 satisfies both conditions, so

$$\mathbb{P}(H \geq 175, W \geq 70) = \frac{1}{5}.$$

The two marginal probabilities do not by themselves determine the joint probability.

2.5.2 Joint cumulative distribution functions and marginals

Definition 2.5.3 (Joint c.d.f.). The joint c.d.f. of (X, Y) is

$$F_{X,Y}(x, y) := \mathbb{P}(X \leq x, Y \leq y).$$

More generally, for $\mathbf{X} = (X_1, \dots, X_d)$,

$$F_{X_1, \dots, X_d}(x_1, \dots, x_d) := \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d).$$

The marginal c.d.f.s are recovered by letting the other coordinate tend to infinity:

$$F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y), \quad F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y).$$

In general, the converse is false: F_X and F_Y do not determine $F_{X,Y}$.

2.5.3 Jointly discrete random variables

Definition 2.5.4 (Joint p.m.f.). If (X, Y) takes values in a finite or countable subset of $\mathbb{R}^2$, its joint p.m.f. is

$$p_{X,Y}(x, y) := \mathbb{P}(X = x, Y = y).$$

The marginal p.m.f.s are obtained by summation:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Moreover,

$$F_{X,Y}(x, y) = \sum_{u \leq x} \sum_{v \leq y} p_{X,Y}(u, v).$$

For a rectangle $[x_1, x_2] \times [y_1, y_2]$, point masses at the boundary are handled by left limits. In particular,

$$\begin{aligned} & \mathbb{P}(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) \\ &= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1^-, y_2) - F_{X,Y}(x_2, y_1^-) + F_{X,Y}(x_1^-, y_1^-). \end{aligned}$$

2.5.4 Jointly continuous random variables

Definition 2.5.5 (Joint p.d.f.). The random variables X and Y are jointly continuous if there is a nonnegative function $f_{X,Y}$ such that

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du.$$

The function $f_{X,Y}$ is the joint p.d.f.

For a sufficiently regular region $B \subseteq \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in B) = \iint_B f_{X,Y}(x, y) \, dx \, dy.$$

The marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx.$$

When the joint c.d.f. is twice differentiable,

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y).$$

Remark 2.5.6 (Continuous marginals need not give a joint density). Let $X \sim \text{Unif}[0, 1]$ and set $Y = X$. Both marginals are continuous, but (X, Y) is concentrated on the line $y = x$, a set of two-dimensional area zero. Hence the pair does not possess an ordinary two-dimensional density.

2.5.5 Expectation of functions of random vectors

Proposition 2.5.7 (Joint LOTUS). For a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$\mathbb{E}[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y) p_{X,Y}(x, y), & \text{jointly discrete,} \\ \int_{\mathbb{R}^2} g(x, y) f_{X,Y}(x, y) \, dx \, dy, & \text{jointly continuous,} \end{cases}$$

whenever the sum or integral is well-defined.

Taking $g(x, y) = x + y$ gives

$$\mathbb{E}(X + Y) = \mathbb{E}X + \mathbb{E}Y.$$

More generally, expectation is linear without any independence assumption.

Proposition 2.5.8 (Linearity of expectation). For random variables $X_1, \dots, X_n$ and constants $a_1, \dots, a_n$,

$$\mathbb{E} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n a_i \mathbb{E} X_i.$$

2.5.6 The coupon collector problem

There are N possible card types. Each draw produces one type, uniformly and independently from the N possibilities.

Example 2.5.9 (Expected number of distinct types after n draws). Let X be the number of distinct types observed in the first n draws. For $i = 1, \dots, N$, let

$$I_i = \mathbf{1}_{\{\text{type } i \text{ appears among the first } n \text{ draws}\}}.$$

Then $X = \sum_{i=1}^N I_i$, and

$$\mathbb{E}I_i = 1 - \mathbb{P}(I_i = 0) = 1 - \left(1 - \frac{1}{N}\right)^n.$$

By linearity,

$$\boxed{\mathbb{E}X = N \left[1 - \left(1 - \frac{1}{N}\right)^n\right].}$$

Example 2.5.10 (Expected time to complete the collection). Let T be the total number of draws needed to see all N types. After i distinct types have already appeared, the probability that the next draw is new is

$$p_i = \frac{N - i}{N}.$$

Let Y_i be the additional number of draws needed to move from i observed types to $i + 1$ observed types. Then

$$Y_i \sim \text{Geom}\left(\frac{N - i}{N}\right), \quad \mathbb{E}Y_i = \frac{N}{N - i},$$

and

$$T = \sum_{i=0}^{N-1} Y_i.$$

Therefore

$$\boxed{\mathbb{E}T = \sum_{i=0}^{N-1} \frac{N}{N - i} = N \sum_{j=1}^N \frac{1}{j}.$$

2.6 Independence, Covariance, and Sums of Random Variables

2.6.1 Independence of random variables

Definition 2.6.1 (Independence of two random variables). Random variables X and Y are independent, written $X \perp\!\!\!\perp Y$, if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

for all suitable sets $A, B \subseteq \mathbb{R}$.

It is enough to test half-lines:

$$X \perp\!\!\!\perp Y \iff F_{X,Y}(x, y) = F_X(x)F_Y(y) \quad \text{for all } x, y \in \mathbb{R}.$$

In the discrete and jointly continuous cases, respectively,

$$X \perp\!\!\!\perp Y \iff p_{X,Y}(x,y) = p_X(x)p_Y(y),$$

and

$$X \perp\!\!\!\perp Y \iff f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad \text{a.e.}$$

More generally, $X_1, \dots, X_n$ are independent if their joint distribution factors into the product of their marginal distributions. Functions of disjoint groups of independent random variables remain independent: if $X_1, \dots, X_n$ are independent, then

$$g(X_1, \dots, X_k) \perp\!\!\!\perp h(X_{k+1}, \dots, X_n)$$

whenever the functions are well-defined.

Proposition 2.6.2 (Factorization of expectation). *If $X \perp\!\!\!\perp Y$ and the expectations exist, then*

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \mathbb{E}[h(Y)].$$

In particular,

$$\mathbb{E}[XY] = \mathbb{E}X \mathbb{E}Y.$$

2.6.2 Covariance and variance of a sum

Definition 2.6.3 (Covariance). For random variables with finite second moments,

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X \mathbb{E}Y.$$

Covariance is symmetric and satisfies

$$\text{Cov}(X, X) = \text{Var}(X).$$

Proposition 2.6.4 (Variance of a sum). *For random variables $X_1, \dots, X_n$ with finite second moments,*

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

If X_i and X_j are independent, then $\text{Cov}(X_i, X_j) = 0$. Consequently, for independent random variables,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

2.6.3 Uncorrelated does not imply independent

Definition 2.6.5 (Uncorrelated random variables). Random variables X and Y are *uncorrelated* if

$$\text{Cov}(X, Y) = 0.$$

Independence implies zero covariance whenever second moments exist, but the converse is false.

Example 2.6.6 (Dependent but uncorrelated). Let

$$\mathbb{P}(X = -1) = \mathbb{P}(X = 0) = \mathbb{P}(X = 1) = \frac{1}{3},$$

and define

$$Y = \mathbf{1}_{\{X=0\}}.$$

Then

$$\mathbb{E}X = 0, \quad \mathbb{E}Y = \frac{1}{3}, \quad \mathbb{E}[XY] = 0,$$

so $\text{Cov}(X, Y) = 0$. Nevertheless, Y is a deterministic function of X , and hence the variables are dependent. For example,

$$\mathbb{P}(X = 0, Y = 0) = 0 \neq \mathbb{P}(X = 0)\mathbb{P}(Y = 0) = \frac{1}{3} \cdot \frac{2}{3}.$$

2.6.4 Distribution of a sum and convolution

Let $Z = X + Y$.

If X and Y are jointly discrete, then

$$p_Z(z) = \sum_x p_{X,Y}(x, z - x).$$

If $X \perp\!\!\!\perp Y$, this becomes the discrete convolution

$$p_{X+Y}(z) = \sum_x p_X(x)p_Y(z - x).$$

If X and Y have a joint density, then

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_{X,Y}(z - y, y) \, dy.$$

Under independence,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(z - y)f_Y(y) \, dy = (f_X * f_Y)(z).$$

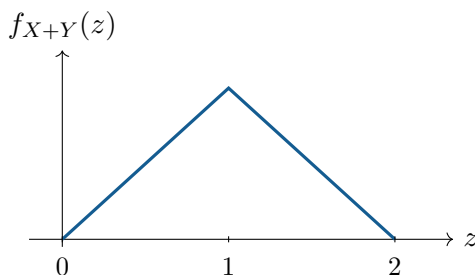
Example 2.6.7 (Sum of two independent uniform variables). Let $X \perp\!\!\!\perp Y$ with $X, Y \sim \text{Unif}[0, 1]$. Since

$$f_X(x) = f_Y(x) = \mathbf{1}_{[0,1]}(x),$$

for $0 \leq z \leq 2$,

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(x)\mathbf{1}_{[0,1]}(z - x) \, dx \\ &= \text{length}([0, 1] \cap [z - 1, z]) \\ &= \begin{cases} z, & 0 \leq z \leq 1, \\ 2 - z, & 1 < z \leq 2. \end{cases} \end{aligned}$$

It is zero outside $[0, 2]$.



Theorem 2.6.8 (Sums of independent normal variables). *If $X_1, \dots, X_n$ are independent and*

$$X_i \sim \mathcal{N}(\mu_i, \sigma_i^2),$$

then

$$\sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right).$$

Convolution proof for two variables. For two centered variables, completing the square in

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_1} e^{-(z-y)^2/(2\sigma_1^2)} \frac{1}{\sqrt{2\pi}\sigma_2} e^{-y^2/(2\sigma_2^2)} dy$$

shows that the result is the density of $\mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$. Affine shifts give the means, and induction gives the general statement. A shorter proof using moment generating functions appears below. $\square$

2.7 Moment Generating Functions

2.7.1 Definition and domain of existence

Definition 2.7.1 (Moment generating function). The *moment generating function*, or *m.g.f.*, of a random variable X is

$$M_X(t) := \mathbb{E}[e^{tX}],$$

for those $t \in \mathbb{R}$ at which the expectation is finite.

For a discrete random variable,

$$M_X(t) = \sum_x e^{tx} p_X(x),$$

and for a continuous random variable,

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx.$$

The m.g.f. need not exist away from $t = 0$. For instance, if $X \sim \text{Exp}(\lambda)$, then $M_X(t)$ is finite only for $t < \lambda$, while the Cauchy distribution has no finite m.g.f. for any $t \neq 0$.

2.7.2 Independent sums

Proposition 2.7.2 (M.g.f. of an independent sum). *If $X \perp\!\!\!\perp Y$ and both sides are finite, then*

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

More generally, for independent $X_1, \dots, X_n$,

$$M_{\sum_i X_i}(t) = \prod_{i=1}^n M_{X_i}(t).$$

Proof. Independence and factorization of expectation give

$$M_{X+Y}(t) = \mathbb{E}[e^{tX} e^{tY}] = \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}].$$

□

Theorem 2.7.3 (Uniqueness of the m.g.f.). *If two random variables have finite m.g.f.s on an open interval containing 0, and those m.g.f.s agree on that interval, then the two random variables have the same distribution.*

2.7.3 Moments from derivatives

Proposition 2.7.4 (Moments via the m.g.f.). *When differentiation under the expectation is justified,*

$$M_X^{(k)}(0) = \mathbb{E}[X^k], \quad k \geq 1.$$

Thus the Taylor expansion of the m.g.f. near zero is

$$M_X(t) = 1 + t\mathbb{E}X + \frac{t^2}{2}\mathbb{E}[X^2] + \dots$$

2.7.4 Examples and applications

Distribution	Moment generating function
$X \sim \text{Bernoulli}(p)$	$M_X(t) = 1 - p + pe^t$
$X \sim \text{Bin}(n, p)$	$M_X(t) = (1 - p + pe^t)^n$
$X \sim \text{Poisson}(\lambda)$	$M_X(t) = \exp\{\lambda(e^t - 1)\}$
$X \sim \text{Exp}(\lambda)$	$M_X(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$
$X \sim \mathcal{N}(\mu, \sigma^2)$	$M_X(t) = \exp\{\mu t + \frac{1}{2}\sigma^2 t^2\}$

Example 2.7.5 (Mean and variance of a Poisson variable). For $X \sim \text{Poisson}(\lambda)$,

$$M_X(t) = e^{\lambda(e^t - 1)}.$$

Hence

$$M'_X(0) = \lambda, \quad M''_X(0) = \lambda + \lambda^2.$$

Therefore

$$\mathbb{E}X = \lambda, \quad \text{Var}(X) = M''_X(0) - [M'_X(0)]^2 = \lambda.$$

Example 2.7.6 (Addition of independent Poisson variables). If

$$X \sim \text{Poisson}(\lambda), \quad Y \sim \text{Poisson}(\mu), \quad X \perp\!\!\!\perp Y,$$

then

$$\begin{aligned} M_{X+Y}(t) &= e^{\lambda(e^t-1)} e^{\mu(e^t-1)} \\ &= e^{(\lambda+\mu)(e^t-1)}. \end{aligned}$$

By uniqueness,

$$X + Y \sim \text{Poisson}(\lambda + \mu).$$

Example 2.7.7 (Addition of independent normal variables). If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ are independent, then

$$\begin{aligned} M_{\sum_i X_i}(t) &= \prod_i \exp\left(\mu_i t + \frac{1}{2} \sigma_i^2 t^2\right) \\ &= \exp\left(\left(\sum_i \mu_i\right) t + \frac{1}{2} \left(\sum_i \sigma_i^2\right) t^2\right), \end{aligned}$$

which is the m.g.f. of the normal law stated earlier.

2.8 Limit Theorems

The preceding sections study finite collections and finite sums of random variables. We now let the number of summands tend to infinity and ask about limiting behavior.

2.8.1 Markov's inequality

Theorem 2.8.1 (Markov's inequality). *If $X \geq 0$ and $a > 0$, then*

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.}$$

Proof. Pointwise,

$$X \geq a \mathbf{1}_{\{X \geq a\}}.$$

Taking expectations gives

$$\mathbb{E}X \geq a \mathbb{P}(X \geq a).$$

□

Two useful consequences are

$$\mathbb{P}(|X| \geq a) \leq \frac{\mathbb{E}|X|}{a}$$

and, for a nonnegative increasing function g on $[0, \infty)$,

$$\mathbb{P}(X \geq a) = \mathbb{P}(g(X) \geq g(a)) \leq \frac{\mathbb{E}[g(X)]}{g(a)}.$$

2.8.2 Chebyshev's inequality

Theorem 2.8.2 (Chebyshev's inequality). *If X has finite mean μ and finite variance, then for every $a > 0$,*

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}.$$

Proof. Apply Markov's inequality to the nonnegative random variable $(X - \mu)^2$. □

2.8.3 Convergence in probability

Definition 2.8.3 (Convergence in probability). A sequence Z_n converges in probability to Z , written

$$Z_n \xrightarrow{\mathbb{P}} Z,$$

if for every $\varepsilon > 0$,

$$\mathbb{P}(|Z_n - Z| > \varepsilon) \longrightarrow 0.$$

2.8.4 Weak law of large numbers

Theorem 2.8.4 (Weak law of large numbers). *Let $X_1, X_2, \dots$ be independent and identically distributed random variables with*

$$\mathbb{E}X_i = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \mu.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \longrightarrow 0.$$

Proof. Independence gives

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

Therefore, by Chebyshev's inequality,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

□

Remark 2.8.5 (Why independence matters). If $X_1 = \dots = X_n = X$, then

$$\frac{1}{n} \sum_{i=1}^n X_i = X$$

and

$$\text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \text{Var}(X),$$

so averaging does not reduce the randomness. The variance reduction in the weak law relies on sufficiently weak dependence, and in the stated theorem on independence.

2.8.5 Central limit theorem

Definition 2.8.6 (Convergence in distribution). We say Z_n converges in distribution to Z , written

$$Z_n \xrightarrow{d} Z,$$

if

$$F_{Z_n}(x) \longrightarrow F_Z(x)$$

for every continuity point x of F_Z .

Theorem 2.8.7 (Central limit theorem). Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{E}X_i = \mu, \quad \text{Var}(X_i) = \sigma^2 \in (0, \infty).$$

Then

$$\boxed{\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{d} \mathcal{N}(0, 1).}$$

Equivalently,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

M.g.f. sketch under an additional local m.g.f. assumption. Assume first that $\mu = 0$ and that the m.g.f. exists near 0. Since

$$\mathbb{E}X_1 = 0, \quad \mathbb{E}X_1^2 = \sigma^2,$$

Taylor expansion gives

$$M_{X_1}(u) = 1 + \frac{\sigma^2 u^2}{2} + o(u^2) \quad (u \rightarrow 0).$$

For

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i,$$

independence yields

$$\begin{aligned} M_{S_n}(t) &= \left[M_{X_1}\left(\frac{t}{\sqrt{n}}\right) \right]^n \\ &= \left[1 + \frac{\sigma^2 t^2}{2n} + o\left(\frac{1}{n}\right) \right]^n \\ &\longrightarrow e^{\sigma^2 t^2/2}, \end{aligned}$$

which is the m.g.f. of $\mathcal{N}(0, \sigma^2)$. Centering handles general μ . □

Remark 2.8.8. The classical central limit theorem itself does not require the existence of an m.g.f. near zero. The argument above is the streamlined m.g.f. proof which is valid under that additional assumption.

Chapter 2 Summary of Key Formulas

CDFs, PMFs, and PDFs

$$F_X(x) = \mathbb{P}(X \leq x), \quad \mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

The left limit satisfies

$$F_X(x^-) = \mathbb{P}(X < x), \quad \mathbb{P}(X = x) = F_X(x) - F_X(x^-).$$

For a discrete random variable,

$$p_X(x) = \mathbb{P}(X = x), \quad F_X(b) = \sum_{x \leq b} p_X(x).$$

For a continuous random variable,

$$F_X(x) = \int_{-\infty}^x f_X(u) \, du, \quad \mathbb{P}(a < X \leq b) = \int_a^b f_X(x) \, dx.$$

Frequently used distributions

$$X \sim \text{Bernoulli}(p) : \quad \mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p,$$

$$X \sim \text{Bin}(n, p) : \quad \mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k},$$

$$X \sim \text{Geom}(p) : \quad \mathbb{P}(X = k) = (1 - p)^{k-1} p, \quad k \geq 1,$$

$$X \sim \text{Poisson}(\lambda) : \quad \mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \geq 0,$$

$$X \sim \text{Unif}[a, b] : \quad f_X(x) = \frac{1}{b - a} \mathbf{1}_{[a, b]}(x),$$

$$X \sim \text{Exp}(\lambda) : \quad f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0, \infty)}(x),$$

$$X \sim \mathcal{N}(\mu, \sigma^2) : \quad f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}.$$

Expectation, variance, and covariance

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx.$$

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}X \mathbb{E}Y,$$

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

For independent variables, all cross-covariance terms vanish.

Joint distributions, independence, and convolution

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

If $X \perp\!\!\!\perp Y$, then

$$F_{X,Y}(x, y) = F_X(x)F_Y(y),$$

and the analogous PMF or PDF factorization holds. For independent variables,

$$p_{X+Y}(z) = \sum_x p_X(x)p_Y(z-x),$$

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(z-y)f_Y(y) \, dy.$$

Transform sampling and Gaussian linear transformations

If $U \sim \text{Unif}[0, 1]$ and F is continuous and strictly increasing, then

$$X = F^{-1}(U)$$

has CDF F . If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Moment generating functions

$$M_X(t) = \mathbb{E}[e^{tX}], \quad X \perp\!\!\!\perp Y \implies M_{X+Y}(t) = M_X(t)M_Y(t).$$

When differentiation is justified,

$$M_X^{(k)}(0) = \mathbb{E}[X^k].$$

In particular,

$$M_{\text{Poisson}(\lambda)}(t) = e^{\lambda(e^t-1)}, \quad M_{\mathcal{N}(\mu, \sigma^2)}(t) = e^{\mu t + \sigma^2 t^2/2}.$$

Inequalities and limit theorems

For $X \geq 0$ and $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

For $\mu = \mathbb{E}X$,

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}.$$

For iid variables with mean μ and variance $\sigma^2 < \infty$,

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \mu,$$

and

$$\frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Chapter 3

Conditional Distributions, Conditional Expectation, and Transformations

Scope of Chapter 3

This chapter extends conditioning from events to random variables. It develops conditional distributions and conditional expectations in both discrete and continuous settings, together with the law of total expectation and applications such as the miner problem, random sums, and conditional Gaussian laws. It also treats the best-prize stopping problem and develops the distributional transformation method for one or several random variables, including inverse images, change of variables, Jacobians, and the pushforward-measure viewpoint.

3.1 Conditional Distributions in the Discrete Case

3.1.1 From conditional probability to a conditional distribution

Recall that, for events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

A conditional distribution applies the same idea when the conditioning event is of the form $\{X = x\}$.

Definition 3.1.1 (Conditional distribution: discrete case). Let X and Y be jointly discrete. For any value x with $p_X(x) = \mathbb{P}(X = x) > 0$, the conditional distribution of Y given $X = x$ is the probability rule

$$A \mapsto \mathbb{P}(Y \in A | X = x) = \frac{\mathbb{P}(Y \in A, X = x)}{\mathbb{P}(X = x)}.$$

The conditional CDF is

$$F_{Y|X}(y | x) := \mathbb{P}(Y \leq y | X = x).$$

For each fixed admissible x , this is an ordinary CDF as a function of y .

The conditional PMF is

$$p_{Y|X}(y | x) := \mathbb{P}(Y = y | X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For each fixed x ,

$$\sum_y p_{Y|X}(y | x) = 1, \quad F_{Y|X}(y | x) = \sum_{v \leq y} p_{Y|X}(v | x).$$

Conversely, at a point y of the support,

$$p_{Y|X}(y | x) = F_{Y|X}(y | x) - F_{Y|X}(y^- | x).$$

Proposition 3.1.2 (Independence and conditional distributions). *If $X \perp\!\!\!\perp Y$, then, whenever $p_X(x) > 0$,*

$$p_{Y|X}(y | x) = p_Y(y), \quad F_{Y|X}(y | x) = F_Y(y).$$

Thus learning the value of X does not change the distribution of Y .

3.1.2 A Poisson conditioning identity

Example 3.1.3 (Two independent Poisson variables conditioned on their sum). Let

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2), \quad X \perp\!\!\!\perp Y.$$

For $0 \leq k \leq n$,

$$\begin{aligned} p_{X|X+Y}(k | n) &= \mathbb{P}(X = k | X + Y = n) \\ &= \frac{\mathbb{P}(X = k, Y = n - k)}{\mathbb{P}(X + Y = n)} \\ &= \frac{e^{-\lambda_1} \lambda_1^k / k! \cdot e^{-\lambda_2} \lambda_2^{n-k} / (n-k)!}{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n / n!} \\ &= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}. \end{aligned}$$

Therefore

$$X | (X + Y = n) \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

3.2 Conditional Expectation and the Law of Total Expectation

3.2.1 Conditioning on a value

Definition 3.2.1 (Conditional expectation given $X = x$: discrete case). Let X and Y be jointly discrete, and suppose $p_X(x) > 0$. The conditional expectation of Y given $X = x$ is

$$\mathbb{E}[Y | X = x] := \sum_y y p_{Y|X}(y | x),$$

provided the sum is absolutely convergent.

As x varies, the conditional expectation is a function of x . Write

$$m(x) := \mathbb{E}[Y | X = x].$$

Definition 3.2.2 (Conditional expectation given a random variable). The conditional expectation of Y given X is the random variable

$$\mathbb{E}[Y | X] := m(X).$$

It is a function of X and therefore contains only information that can be read from X .

For a general event A with $\mathbb{P}(A) > 0$, the same construction gives

$$\mathbb{E}[Y \mid A] = \sum_y y \mathbb{P}(Y = y \mid A)$$

in the discrete case. In the present chapter the most frequent event is $A = \{X = x\}$.

Example 3.2.3 (The Poisson example revisited). From the preceding conditional binomial law,

$$\mathbb{E}[X \mid X + Y = n] = n \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Consequently,

$$\mathbb{E}[X \mid X + Y] = \frac{\lambda_1}{\lambda_1 + \lambda_2} (X + Y).$$

3.2.2 Conditional expectation as a best prediction

Expectation is the best constant prediction in mean square. If $c \in \mathbb{R}$, then

$$\mathbb{E}[(Y - c)^2] = \text{Var}(Y) + (\mathbb{E}Y - c)^2,$$

so the minimizing constant is $c = \mathbb{E}Y$.

Conditional expectation gives the corresponding best prediction when the information in X is available.

Proposition 3.2.4 (Mean-square prediction). *Suppose Y is square-integrable and let $m(X) = \mathbb{E}[Y \mid X]$. For every square-integrable function $h(X)$,*

$$\mathbb{E}[(Y - h(X))^2] = \mathbb{E}[(Y - m(X))^2] + \mathbb{E}[(m(X) - h(X))^2].$$

Hence $h(X) = \mathbb{E}[Y \mid X]$ minimizes the mean-square prediction error among all functions of X .

Proof. Expand

$$Y - h(X) = (Y - m(X)) + (m(X) - h(X)).$$

The cross term vanishes because

$$\mathbb{E}[Y - m(X) \mid X] = 0.$$

Taking expectations gives the stated decomposition. $\square$

Example 3.2.5 (A die roll and partial information). Roll a fair die. Let Y be the number shown and let

$$X = \mathbf{1}_{\{Y \text{ is even}\}}.$$

Without any extra information, the best constant prediction is

$$\mathbb{E}Y = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2}.$$

If we are told whether the outcome is even or odd, then

$$\mathbb{E}[Y \mid X = 1] = \frac{2 + 4 + 6}{3} = 4, \quad \mathbb{E}[Y \mid X = 0] = \frac{1 + 3 + 5}{3} = 3.$$

Therefore

$$\mathbb{E}[Y \mid X] = 4\mathbf{1}_{\{X=1\}} + 3\mathbf{1}_{\{X=0\}} = 3 + X.$$

Two immediate identities are

$$\mathbb{E}[X \mid X] = X$$

and, if $X \perp\!\!\!\perp Y$,

$$\mathbb{E}[X \mid Y] = \mathbb{E}X.$$

3.2.3 The law of total expectation

Theorem 3.2.6 (Law of total expectation). *Whenever the expectations exist,*

$$\mathbb{E}Y = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

In the discrete case,

$$\mathbb{E}Y = \sum_x \mathbb{E}[Y \mid X = x]p_X(x).$$

Proof in the discrete case. Using $p_{X,Y}(x, y) = p_{Y|X}(y \mid x)p_X(x)$,

$$\begin{aligned} \sum_x \mathbb{E}[Y \mid X = x]p_X(x) &= \sum_x \sum_y y p_{Y|X}(y \mid x)p_X(x) \\ &= \sum_y y \sum_x p_{X,Y}(x, y) \\ &= \sum_y y p_Y(y) \\ &= \mathbb{E}Y. \end{aligned}$$

□

3.2.4 Examples based on total expectation

Example 3.2.7 (The miner problem). A miner is trapped in a mine with three tunnels. Each time a tunnel is chosen uniformly at random.

- Tunnel 1 leads to safety after 3 hours.
- Tunnel 2 returns the miner to the starting point after 5 hours.
- Tunnel 3 returns the miner to the starting point after 7 hours.

Let T be the total time needed to reach safety, and let J be the first tunnel selected. By the restart structure,

$$\mathbb{E}[T \mid J = 1] = 3, \quad \mathbb{E}[T \mid J = 2] = 5 + \mathbb{E}T, \quad \mathbb{E}[T \mid J = 3] = 7 + \mathbb{E}T.$$

Therefore

$$\mathbb{E}T = \frac{1}{3}(3 + (5 + \mathbb{E}T) + (7 + \mathbb{E}T)),$$

which yields

$$\mathbb{E}T = 15 \text{ hours.}$$

Proposition 3.2.8 (Expectation of a random sum). *Let N be a nonnegative integer-valued random variable, and let $X_1, X_2, \dots$ be iid random variables. Assume N is independent of all the X_i , and that $\mathbb{E}N < \infty$ and $\mathbb{E}|X_1| < \infty$. With*

$$S_N := \sum_{i=1}^N X_i, \quad S_0 := 0,$$

we have

$$\boxed{\mathbb{E}S_N = (\mathbb{E}N)(\mathbb{E}X_1).}$$

Proof. Conditioning on N gives

$$\mathbb{E}[S_N \mid N = n] = n\mathbb{E}X_1.$$

Hence

$$\mathbb{E}S_N = \mathbb{E}[\mathbb{E}[S_N \mid N]] = \mathbb{E}[N\mathbb{E}X_1] = (\mathbb{E}N)(\mathbb{E}X_1).$$

□

3.3 Conditional Distributions and Expectations in the Continuous Case

3.3.1 The density-ratio formula

Suppose (X, Y) is jointly continuous with joint PDF $f_{X,Y}$. Since $\mathbb{P}(X = x) = 0$, the elementary ratio

$$\frac{\mathbb{P}(Y \leq y, X = x)}{\mathbb{P}(X = x)}$$

is not meaningful. We can justify the definition by conditioning on a short interval. For $\Delta > 0$,

$$\begin{aligned} & \mathbb{P}(Y \leq y \mid x \leq X \leq x + \Delta) \\ &= \frac{\int_x^{x+\Delta} \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du}{\int_x^{x+\Delta} f_X(u) \, du}. \end{aligned}$$

As $\Delta \downarrow 0$, this ratio approaches

$$\frac{\int_{-\infty}^y f_{X,Y}(x, v) \, dv}{f_X(x)},$$

provided $f_X(x) > 0$ and the relevant functions are regular enough.

Definition 3.3.1 (Conditional CDF and PDF: continuous case). For x such that $f_X(x) > 0$, define

$$\boxed{F_{Y|X}(y \mid x) := \int_{-\infty}^y \frac{f_{X,Y}(x, v)}{f_X(x)} \, dv}$$

and

$$\boxed{f_{Y|X}(y \mid x) := \frac{f_{X,Y}(x, y)}{f_X(x)}}.$$

For a measurable set A ,

$$\mathbb{P}(Y \in A \mid X = x) = \int_A f_{Y|X}(y \mid x) \, dy.$$

The set may depend on x . For example, if $c > x^2$, then

$$\mathbb{P}(X^2 + Y^2 \leq c \mid X = x) = \int_{-\sqrt{c-x^2}}^{\sqrt{c-x^2}} f_{Y|X}(y \mid x) \, dy.$$

Integrating this conditional probability against $f_X(x)$ recovers the unconditional probability.

3.3.2 Conditional expectation in the continuous case

Definition 3.3.2 (Conditional expectation: continuous case). For $f_X(x) > 0$, define

$$\mathbb{E}[Y \mid X = x] := \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x) \, dy.$$

If $m(x) = \mathbb{E}[Y \mid X = x]$, then

$$\mathbb{E}[Y \mid X] = m(X).$$

The law of total expectation becomes

$$\mathbb{E}Y = \mathbb{E}[\mathbb{E}[Y \mid X]] = \int_{-\infty}^{\infty} \mathbb{E}[Y \mid X = x] f_X(x) \, dx.$$

Conditioning is also useful for probabilities. If A is an event and Y is discrete,

$$\mathbb{P}(A) = \sum_y \mathbb{P}(A \mid Y = y) p_Y(y),$$

while in the continuous case,

$$\mathbb{P}(A) = \int_{-\infty}^{\infty} \mathbb{P}(A \mid Y = y) f_Y(y) \, dy.$$

To compute a variance by conditioning, one may use

$$\text{Var}(X) = \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - (\mathbb{E}[\mathbb{E}[X \mid Y]])^2.$$

This is simply $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ together with the law of total expectation.

3.3.3 The bivariate normal distribution

Definition 3.3.3 (Bivariate normal density). We say that (X, Y) is bivariate normal with means μ_X, μ_Y , standard deviations $\sigma_X, \sigma_Y > 0$, and correlation parameter $-1 < \rho < 1$ if

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left\{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X}\right) \left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right]\right\}.$$

The marginal distributions are

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2).$$

Within the bivariate normal family, X and Y are independent exactly when $\rho = 0$.

Completing the square in x gives the conditional law.

Proposition 3.3.4 (Conditional distribution in a bivariate normal model). *For every y with $f_Y(y) > 0$,*

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2)\right).$$

Consequently,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y}(Y - \mu_Y).$$

As $\rho \rightarrow 1$, the conditional variance tends to zero and the conditional mean approaches the linear relation

$$X \approx \mu_X + \frac{\sigma_X}{\sigma_Y}(Y - \mu_Y).$$

3.4 An Optimal-Stopping Application: The Best-Prize Problem

Suppose n distinct prizes are presented one at a time in uniformly random order. Only one prize may be accepted, and a rejected prize cannot be recalled. Consider the strategy:

1. Reject the first k prizes.
2. Afterward, accept the first prize that is better than all of the first k prizes.

Let B be the event that this strategy selects the overall best prize, and let I be the true position of the best prize. Since the order is random,

$$\mathbb{P}(I = i) = \frac{1}{n}, \quad i = 1, \dots, n.$$

If $i \leq k$, the best prize is rejected, so

$$\mathbb{P}(B \mid I = i) = 0.$$

If $i > k$, the strategy succeeds exactly when the best among the first $i - 1$ prizes lies among the first k positions. Hence

$$\mathbb{P}(B \mid I = i) = \frac{k}{i - 1}.$$

By conditioning on I ,

$$\mathbb{P}_k(B) = \frac{k}{n} \sum_{i=k+1}^n \frac{1}{i - 1} = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j}.$$

For large n ,

$$\mathbb{P}_k(B) \approx \frac{k}{n} \int_k^n \frac{dx}{x} = \frac{k}{n} \log\left(\frac{n}{k}\right).$$

Writing $x = k/n$, the approximation is $x \log(1/x)$, which is maximized at $x = e^{-1}$. Thus the asymptotically optimal strategy is

$$k \approx \frac{n}{e},$$

and the corresponding success probability is approximately

$$\frac{1}{e}.$$

3.5 Functions and Transformations of Random Variables

Chapter 2 explained how to compute $\mathbb{E}[g(X)]$ directly from the distribution of X . We now ask for the full distribution of $g(X)$ and, more generally, of transformations of random vectors.

3.5.1 The preimage method

For a function $g : \mathbb{R} \rightarrow \mathbb{R}$ and a set $A \subseteq \mathbb{R}$, define the preimage

$$g^{-1}(A) := \{x \in \mathbb{R} : g(x) \in A\}.$$

If $Y = g(X)$, then

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \in g^{-1}((-\infty, y])).$$

When X has PDF f_X ,

$$F_Y(y) = \int_{g^{-1}((-\infty, y])} f_X(x) \, dx.$$

This method works even when g is not one-to-one.

Example 3.5.1 (The square of a standard normal variable). Let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$. For $y < 0$, $F_Y(y) = 0$. For $y \geq 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) \\ &= 2\Phi(\sqrt{y}) - 1, \end{aligned}$$

where Φ is the standard normal CDF. Differentiating gives, for $y > 0$,

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

This is the χ^2 density with one degree of freedom.

3.5.2 A strictly monotone transformation

Proposition 3.5.2 (One-dimensional change of variables). *Let $Y = g(X)$, where g is strictly monotone and differentiable on an interval containing the support of X . Suppose the inverse function g^{-1} is differentiable. Then, for y in the range of g ,*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Outside the range of g , $f_Y(y) = 0$.

Proof. If g is increasing, then

$$F_Y(y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)),$$

and differentiation gives the formula. If g is decreasing, then

$$F_Y(y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)),$$

and the absolute value accounts for the negative derivative. □

3.5.3 Two-dimensional transformations and Jacobians

Let (X, Y) have joint PDF $f_{X,Y}$, and define

$$U = g(X, Y), \quad V = h(X, Y).$$

Assume that the transformation is one-to-one on the relevant region, so that

$$X = a(U, V), \quad Y = b(U, V),$$

and assume the partial derivatives are continuous. Define the forward Jacobian

$$J_T(x, y) := \det \begin{pmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{pmatrix}.$$

If $J_T \neq 0$, then

$$f_{U,V}(u, v) = f_{X,Y}(a(u, v), b(u, v)) \frac{1}{|J_T(a(u, v), b(u, v))|}.$$

Equivalently,

$$f_{U,V}(u, v) = f_{X,Y}(a(u, v), b(u, v)) \left| \det \frac{\partial(a, b)}{\partial(u, v)} \right|.$$

Example 3.5.3 (Sum and difference). Let

$$Y_1 = X_1 + X_2, \quad Y_2 = X_1 - X_2.$$

The inverse transformation is

$$X_1 = \frac{Y_1 + Y_2}{2}, \quad X_2 = \frac{Y_1 - Y_2}{2}.$$

The forward Jacobian is

$$\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2,$$

so

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{2} f_{X_1, X_2} \left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2} \right).$$

Example 3.5.4 (Sum and difference of two independent standard normals). Let X_1, X_2 be independent $\mathcal{N}(0, 1)$ variables. Then

$$\begin{aligned} f_{Y_1, Y_2}(y_1, y_2) &= \frac{1}{4\pi} \exp \left(-\frac{y_1^2 + y_2^2}{4} \right) \\ &= \left(\frac{1}{\sqrt{4\pi}} e^{-y_1^2/4} \right) \left(\frac{1}{\sqrt{4\pi}} e^{-y_2^2/4} \right). \end{aligned}$$

Thus

$$Y_1 \perp\!\!\!\perp Y_2, \quad Y_1, Y_2 \sim \mathcal{N}(0, 2).$$

Remark 3.5.5 (Orthogonal transformations of a standard Gaussian vector). Let X_1, X_2 be iid $\mathcal{N}(0, 1)$, let O be a 2×2 orthogonal matrix, and let

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = rO \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}.$$

Then Z_1 and Z_2 are independent and each has distribution $\mathcal{N}(0, r^2)$. The sum–difference example corresponds, after rescaling, to a particular orthogonal matrix.

Example 3.5.6 (Transforming a uniform square into a diamond). Let X_1, X_2 be independent $\text{Unif}[0, 1]$ variables, and again set

$$Y_1 = X_1 + X_2, \quad Y_2 = X_1 - X_2.$$

Since $f_{X_1, X_2} = 1$ on the unit square and 0 elsewhere,

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{2}$$

when

$$0 \leq y_1 + y_2 \leq 2, \quad 0 \leq y_1 - y_2 \leq 2,$$

and the density is 0 otherwise. The support is the diamond with vertices

$$(0, 0), \quad (1, 1), \quad (2, 0), \quad (1, -1).$$

3.5.4 The distribution as a pushforward measure

The distribution of a random variable can be described without choosing a PMF or PDF. If

$$X : (S, \mathcal{F}, \mathbb{P}) \longrightarrow \mathbb{R}$$

is a random variable, define, for a Borel set $B \subseteq \mathbb{R}$,

$$\mu_X(B) := \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)).$$

The probability measure $\mu_X = \mathbb{P} \circ X^{-1}$ is called the distribution, or pushforward measure, of X . In particular,

$$F_X(x) = \mu_X((-\infty, x]).$$

This viewpoint summarizes the relation

$$(S, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X).$$

Chapter 3 Summary of Key Formulas

Conditional laws: discrete case

For $p_X(x) > 0$,

$$p_{Y|X}(y | x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \quad F_{Y|X}(y | x) = \sum_{v \leq y} p_{Y|X}(v | x).$$

If $X \perp\!\!\!\perp Y$, then

$$p_{Y|X}(y | x) = p_Y(y).$$

For independent $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$,

$$X | (X + Y = n) \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

Conditional laws: continuous case

For $f_X(x) > 0$,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \quad F_{Y|X}(y | x) = \int_{-\infty}^y f_{Y|X}(v | x) dv.$$

For a measurable set A ,

$$\mathbb{P}(Y \in A | X = x) = \int_A f_{Y|X}(y | x) dy.$$

Conditional expectation and conditioning identities

$$\mathbb{E}[Y | X = x] = \sum_y y p_{Y|X}(y | x) \quad \text{or} \quad \mathbb{E}[Y | X = x] = \int y f_{Y|X}(y | x) dy.$$

If $m(x) = \mathbb{E}[Y | X = x]$, then

$$\mathbb{E}[Y | X] = m(X), \quad \mathbb{E}Y = \mathbb{E}[\mathbb{E}[Y | X]].$$

Also,

$$\mathbb{E}[X | X] = X, \quad X \perp\!\!\!\perp Y \implies \mathbb{E}[X | Y] = \mathbb{E}X.$$

For an event A ,

$$\mathbb{P}(A) = \sum_y \mathbb{P}(A | Y = y) p_Y(y) \quad \text{or} \quad \mathbb{P}(A) = \int \mathbb{P}(A | Y = y) f_Y(y) dy.$$

Random sums and mean-square prediction

If N is independent of iid $X_1, X_2, \dots$, then

$$\mathbb{E} \left[\sum_{i=1}^N X_i \right] = (\mathbb{E}N)(\mathbb{E}X_1).$$

For any square-integrable function $h(X)$,

$$\mathbb{E}[(Y - h(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - h(X))^2].$$

Thus $\mathbb{E}[Y | X]$ is the best mean-square predictor of Y based on X .

Bivariate normal conditioning

If (X, Y) is bivariate normal with correlation ρ , then

$$X | (Y = y) \sim \mathcal{N} \left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y), \sigma_X^2 (1 - \rho^2) \right),$$

and

$$\mathbb{E}[X | Y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y).$$

Transformations

For $Y = g(X)$,

$$F_Y(y) = \mathbb{P}(X \in g^{-1}((-\infty, y])).$$

If g is strictly monotone and differentiable,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

For a one-to-one transformation $(U, V) = T(X, Y)$,

$$f_{U,V}(u, v) = f_{X,Y}(T^{-1}(u, v)) \left| \det DT^{-1}(u, v) \right|.$$

The distribution of X as a measure is

$$\mu_X(B) = \mathbb{P}(X^{-1}(B)) = (\mathbb{P} \circ X^{-1})(B).$$

Best-prize strategy

If the first k of n prizes are rejected and the first later record is accepted, then

$$\mathbb{P}_k(\text{select best}) = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j} \approx \frac{k}{n} \log \left(\frac{n}{k} \right).$$

The large- n optimum is

$$k \approx \frac{n}{e}, \quad \mathbb{P}_k(\text{select best}) \approx \frac{1}{e}.$$

Chapter 4

Discrete-Time Markov Chains

Scope of Chapter 4

This chapter develops discrete-time, discrete-state Markov chains, including transition matrices, the Chapman–Kolmogorov equations, communicating classes, recurrence and transience, stationary and limiting distributions, periodicity, positive recurrence, long-run averages, and gambler’s ruin. It then studies occupation times in transient states, the Galton–Watson branching process, probability generating functions, extinction probabilities, and first-return generating functions for one-dimensional random walk.

4.1 Stochastic Processes and the Markov Property

4.1.1 Stochastic processes

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let S be a state space.

Definition 4.1.1 (Stochastic process). A stochastic process indexed by a set T is a collection of random variables

$$\{X_t : t \in T\}, \quad X_t : \Omega \longrightarrow S.$$

The set T is called the *index set* or *time domain*, and S is called the *state space*.

The two features of a process may be classified separately:

- the process is in *discrete time* if T is discrete, such as $T = \mathbb{N}_0 := \{0, 1, 2, \dots\}$;
- it is in *continuous time* if T is an interval of $\mathbb{R}$;
- it is *discrete-valued* if S is finite or countable;
- it is *continuous-valued* if S is a continuum, usually a subset of $\mathbb{R}^d$.

Typical examples are:

process	time	state space
random walk	discrete	discrete
Poisson process	continuous	discrete
Brownian motion	continuous	continuous

In this chapter we consider only discrete-time, discrete-state processes, usually with $T = \mathbb{N}_0$ and a finite or countable state space S . If $X_n = i$, we say that the process is *in state i at time n* .

4.1.2 The Markov property

The defining feature of a Markov chain is that, once the present state is known, the past gives no further information about the next state.

Definition 4.1.2 (Markov chain). A discrete-time process $\{X_n\}_{n \geq 0}$ with state space $\mathbf{S}$ is a Markov chain if, whenever the conditioning event has positive probability,

$$\begin{aligned} \mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \\ = \mathbb{P}(X_{n+1} = j \mid X_n = i) \end{aligned}$$

for every $n \geq 0$ and all states $i_0, \dots, i_{n-1}, i, j$.

Definition 4.1.3 (Time-homogeneous chain and one-step transition probabilities). A Markov chain is *time homogeneous* if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i)$$

does not depend on n . In that case we write

$$p_{ij} := \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

and call p_{ij} the one-step transition probability from i to j .

Unless stated otherwise, all Markov chains in this chapter are time homogeneous.

4.1.3 Basic examples

Example 4.1.4 (A general random walk). Let $\xi_1, \xi_2, \dots$ be iid integer-valued random variables with PMF

$$q(r) = \mathbb{P}(\xi_1 = r), \quad r \in \mathbb{Z},$$

and let X_0 be fixed. Define

$$X_n = X_0 + \sum_{k=1}^n \xi_k.$$

Then

$$\begin{aligned} p_{ij} &= \mathbb{P}(X_{n+1} = j \mid X_n = i) \\ &= \mathbb{P}(i + \xi_{n+1} = j) \\ &= q(j - i). \end{aligned}$$

Thus the distribution of the next state depends on the present state only through the required increment $j - i$.

Example 4.1.5 (Branching process). Let $\xi_{n,m}$ denote the number of offspring of the m th individual in generation n , and suppose all the variables $\{\xi_{n,m}\}$ are iid nonnegative integer-valued random variables. If X_n is the number of individuals in generation n , then

$$X_{n+1} = \sum_{m=1}^{X_n} \xi_{n,m},$$

where an empty sum is 0. Conditional on $X_n = i$,

$$p_{ij} = \mathbb{P}\left(\sum_{m=1}^i \xi_{n,m} = j\right),$$

which is independent of all generations before n . Hence $\{X_n\}$ is a Markov chain. Notice that state 0 is absorbing.

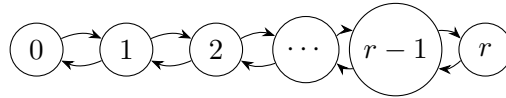
Example 4.1.6 (Ehrenfest chain). There are r balls divided between two urns. At each step, one of the r balls is chosen uniformly and moved to the other urn. Let

X_n = number of balls in the first urn after step n .

If $X_n = k$, then $r - k$ balls are in the second urn and k balls are in the first. Therefore

$$p_{k,k+1} = \frac{r-k}{r}, \quad p_{k,k-1} = \frac{k}{r}, \quad 0 \leq k \leq r,$$

with all other transition probabilities equal to 0.



The transition probabilities vary with the current state, but not with time.

4.2 Transition Matrices and Multi-Step Transitions

4.2.1 The transition matrix

For every state i ,

$$p_{ij} \geq 0, \quad \sum_{j \in S} p_{ij} = 1.$$

Thus each row of transition probabilities is itself a probability distribution.

Definition 4.2.1 (Transition matrix). If $S = \{0, 1, \dots, m-1\}$ is finite, the matrix

$$P = (p_{ij})_{0 \leq i, j \leq m-1}$$

is called the transition matrix. It is a *row-stochastic matrix*: its entries are nonnegative and every row sums to 1.

Example 4.2.2 (A two-state phone model). Let state 0 mean that a customer uses an iPhone and state 1 mean that the customer uses an Android phone. Consider

$$P = \begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix}.$$

Here α is the probability that an iPhone user remains an iPhone user for the next period, while β is the probability that an Android user switches to an iPhone.

The numerical example in the notes is

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}.$$

4.2.2 The Chapman–Kolmogorov equations

Definition 4.2.3 (n -step transition probabilities). For $n \geq 0$, define

$$p_{ij}^{(n)} := \mathbb{P}(X_{m+n} = j \mid X_m = i).$$

Time homogeneity makes this quantity independent of m . By convention,

$$p_{ij}^{(0)} = \mathbf{1}_{\{i=j\}}.$$

The superscript (n) is an index indicating the number of steps; it is not an ordinary power of the scalar p_{ij} .

Theorem 4.2.4 (Chapman–Kolmogorov equations). For all $m, n \geq 0$ and states i, j ,

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

Proof. Condition on the state at the intermediate time m :

$$\begin{aligned} p_{ij}^{(m+n)} &= \mathbb{P}(X_{m+n} = j \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j, X_m = k \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j \mid X_m = k, X_0 = i) \mathbb{P}(X_m = k \mid X_0 = i). \end{aligned}$$

By the Markov property and time homogeneity, the two factors are $p_{kj}^{(n)}$ and $p_{ik}^{(m)}$, respectively. $\square$

For a finite state space, matrix multiplication gives the same formula:

$$P^{m+n} = P^m P^n, \quad p_{ij}^{(n)} = (P^n)_{ij}.$$

Example 4.2.5 (Several steps in the phone model). For

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix},$$

direct multiplication gives

$$P^4 = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}$$

and

$$P^8 \approx \begin{pmatrix} 0.5714567 & 0.4285433 \\ 0.5713911 & 0.4286089 \end{pmatrix}.$$

The two rows are becoming almost identical. This suggests that, after many transitions, the distribution may become nearly independent of the initial state. Later we state conditions under which this convergence is valid.

4.2.3 Evolution of an initial distribution

Let

$$\mu_n(j) = \mathbb{P}(X_n = j)$$

and regard μ_n as a row vector. The law of total probability yields

$$\mu_{n+1}(j) = \sum_i \mu_n(i) p_{ij},$$

so

$$\boxed{\mu_{n+1} = \mu_n P, \quad \mu_n = \mu_0 P^n.}$$

For example, if $\mu_0 = (0.4, 0.6)$ in the numerical phone model, then

$$\mu_4 = \mu_0 P^4 = (0.57004, 0.42996),$$

which is already close to the long-run distribution found later in the chapter.

4.2.4 Two nearest-neighbor chains

Example 4.2.6 (Nearest-neighbor random walk on $\mathbb{Z}$). For a fixed $p \in (0, 1)$ and $q = 1 - p$, let

$$p_{i,i+1} = p, \quad p_{i,i-1} = q, \quad i \in \mathbb{Z}.$$

Every step moves one unit to the right or one unit to the left.

Example 4.2.7 (A random walk with absorbing boundaries). Fix $N \geq 1$ and take state space $\{0, 1, \dots, N\}$. For $1 \leq i \leq N - 1$, let

$$p_{i,i+1} = p, \quad p_{i,i-1} = q,$$

and impose the boundary conditions

$$p_{00} = p_{NN} = 1.$$

Once the chain reaches 0 or N , it stays there. This is the Markov-chain model underlying the gambler's ruin problem.

4.3 Communicating Classes and Irreducibility

A directed graph is often the most convenient way to display the possible transitions. Each state is represented by a vertex, and an arrow $i \rightarrow j$ is drawn whenever $p_{ij} > 0$.

4.3.1 Accessibility and communication

Definition 4.3.1 (Accessibility). State j is *accessible from* state i , written

$$i \longrightarrow j,$$

if there exists $n \geq 0$ such that

$$p_{ij}^{(n)} > 0.$$

Thus there is a path of positive probability from i to j in some finite number of steps.

Definition 4.3.2 (Communication). States i and j *communicate*, written

$$i \longleftrightarrow j,$$

if both $i \rightarrow j$ and $j \rightarrow i$.

Proposition 4.3.3 (Communication is an equivalence relation). *Communication is reflexive, symmetric, and transitive. Consequently, it partitions the state space into disjoint communicating classes.*

Proof. Reflexivity follows from $p_{ii}^{(0)} = 1$, and symmetry is part of the definition. For transitivity, suppose $i \leftrightarrow j$ and $j \leftrightarrow k$. Choose $m, n \geq 0$ with $p_{ij}^{(m)} > 0$ and $p_{jk}^{(n)} > 0$. By Chapman–Kolmogorov,

$$p_{ik}^{(m+n)} = \sum_{\ell} p_{i\ell}^{(m)} p_{\ell k}^{(n)} \geq p_{ij}^{(m)} p_{jk}^{(n)} > 0,$$

so $i \rightarrow k$. The reverse accessibility follows similarly, and hence $i \leftrightarrow k$. $\square$

Definition 4.3.4 (Irreducible chain). A Markov chain is *irreducible* if its state space consists of a single communicating class; equivalently, every state is accessible from every other state.

Example 4.3.5 (A transition graph). Consider

$$P = \begin{pmatrix} 0.2 & 0 & 0.1 & 0.7 \\ 0 & 0 & 0.5 & 0.5 \\ 0.8 & 0.1 & 0.1 & 0 \\ 0 & 0 & 0.6 & 0.4 \end{pmatrix}.$$

Although several one-step transitions are impossible, every state can reach every other state through a path of positive probability. Thus the chain is irreducible.

Example 4.3.6 (Several communicating classes). A chain may have, for example, three communicating classes $\{0, 1\}$, $\{2\}$, and $\{3\}$. A possible transition graph is



The one-way arrows between classes do not merge them: communication requires paths in both directions.

4.4 Recurrence and Transience

From now on, write $\mathbb{P}_i$ and $\mathbb{E}_i$ for probability and expectation conditional on $X_0 = i$.

4.4.1 First visits and first returns

For a state j , define the first positive hitting time

$$T_j^+ := \inf\{n \geq 1 : X_n = j\},$$

with the convention $\inf \emptyset = \infty$. Set

$$f_{ij} := \mathbb{P}_i(T_j^+ < \infty).$$

If $i \neq j$, this is the probability of ever visiting j when the chain starts from i . If $i = j$, it is the probability of returning to i after time 0.

Definition 4.4.1 (Recurrent and transient states). A state i is

- *recurrent* if $f_{ii} = 1$;
- *transient* if $f_{ii} < 1$.

Thus a recurrent state is revisited with probability one, whereas a transient state has positive probability of being left forever.

4.4.2 The number of visits

Let

$$N_j := \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=j\}}$$

be the total number of visits to j , including time 0 if $X_0 = j$.

Proposition 4.4.2 (Visits to a transient state). *If j is transient and the chain starts from j , then*

$$\mathbb{P}_j(N_j = n) = f_{jj}^{n-1}(1 - f_{jj}), \quad n = 1, 2, \dots,$$

and consequently

$$\mathbb{E}_j N_j = \frac{1}{1 - f_{jj}} < \infty.$$

If $i \neq j$, then

$$\mathbb{E}_i N_j = \frac{f_{ij}}{1 - f_{jj}}.$$

Proof. After each visit to j , the Markov property restarts the same return experiment. Each further return occurs with probability f_{jj} , so the number of visits is geometric. Starting from $i \neq j$, the chain first reaches j with probability f_{ij} and then begins the same geometric sequence of visits. $\square$

The occupation-count identity gives a particularly useful criterion. Note that,

$$\begin{aligned} \mathbb{E}_i N_j &= \mathbb{E}_i \left[\sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=j\}} \right] \\ &= \sum_{n=0}^{\infty} \mathbb{P}_i(X_n = j) \\ &= \sum_{n=0}^{\infty} p_{ij}^{(n)}. \end{aligned}$$

Theorem 4.4.3 (Recurrence criterion). *For every state i ,*

$$i \text{ is recurrent} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty,$$

and

$$i \text{ is transient} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty.$$

4.4.3 Recurrence is a class property

Proposition 4.4.4 (Propagation of recurrence). *If i is recurrent and $i \leftrightarrow j$, then j is recurrent. Consequently, all states in a communicating class are either all recurrent or all transient.*

Proof. Since $i \leftrightarrow j$, choose $k, m \geq 0$ such that

$$p_{ji}^{(m)} > 0, \quad p_{ij}^{(k)} > 0.$$

For every $n \geq 0$, Chapman–Kolmogorov gives

$$p_{jj}^{(m+n+k)} \geq p_{ji}^{(m)} p_{ii}^{(n)} p_{ij}^{(k)}.$$

Summing over n and using recurrence of i ,

$$\sum_{r=0}^{\infty} p_{jj}^{(r)} \geq p_{ji}^{(m)} p_{ij}^{(k)} \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

Hence j is recurrent. Applying the result in the reverse direction shows that recurrence, and therefore transience, is constant on each communicating class. $\square$

A useful one-sided consequence is the following: if i is recurrent and $i \rightarrow j$, then necessarily $j \rightarrow i$ and j is recurrent. A recurrent class cannot be left without a possible return.

4.5 Recurrence of Random Walks

The recurrence criterion can be applied by estimating return probabilities.

4.5.1 One-dimensional nearest-neighbor random walk

Consider the random walk on $\mathbb{Z}$ with right-step probability p and left-step probability $q = 1 - p$. The chain is irreducible, so it is enough to classify state 0.

A return to 0 is possible only after an even number of steps. In $2n$ steps the walk must make exactly n right moves and n left moves, so

$$p_{00}^{(2n)} = \binom{2n}{n} p^n q^n, \quad p_{00}^{(2n+1)} = 0.$$

Stirling's approximation gives

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}},$$

and therefore

$$p_{00}^{(2n)} \sim \frac{[4pq]^n}{\sqrt{\pi n}}.$$

Since $4pq \leq 1$, with equality exactly when $p = q = 1/2$, we obtain:

- if $p = 1/2$, then $p_{00}^{(2n)} \asymp n^{-1/2}$ and $\sum_n p_{00}^{(n)} = \infty$;
- if $p \neq 1/2$, then $4pq < 1$ and $\sum_n p_{00}^{(n)} < \infty$.

Hence the symmetric walk is recurrent, while every biased nearest-neighbor walk on $\mathbb{Z}$ is transient.

4.5.2 Symmetric random walks in dimensions two and three

For the simple symmetric random walk on $\mathbb{Z}^2$, the return probability satisfies

$$p_{00}^{(2n)} \asymp \frac{1}{n}.$$

Thus

$$\sum_{n=0}^{\infty} p_{00}^{(n)} = \infty,$$

so the two-dimensional walk is recurrent.

For the simple symmetric random walk on $\mathbb{Z}^3$,

$$p_{00}^{(2n)} \asymp n^{-3/2}.$$

The corresponding series converges, so the three-dimensional walk is transient.

Recurrence classification of simple symmetric random walk

For the simple symmetric nearest-neighbor random walk on $\mathbb{Z}^d$,

$$\begin{array}{ll} d = 1, 2 & : \text{ recurrent,} \\ d = 3 & : \text{ transient.} \end{array}$$

4.5.3 Finite irreducible chains

Theorem 4.5.1. *Every state of a finite irreducible Markov chain is recurrent.*

Proof. By the class property, it is enough to prove that at least one state is recurrent. Suppose instead that every state is transient. Fix an initial state i . For each state j , transience implies

$$\mathbb{E}_i N_j = \sum_{n=0}^{\infty} p_{ij}^{(n)} < \infty.$$

Because the state space is finite,

$$\sum_{j \in S} \mathbb{E}_i N_j < \infty.$$

On the other hand, exactly one state is occupied at every time, so

$$\begin{aligned} \sum_{j \in S} N_j &= \sum_{j \in S} \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=j\}} \\ &= \sum_{n=0}^{\infty} \sum_{j \in S} \mathbf{1}_{\{X_n=j\}} \\ &= \sum_{n=0}^{\infty} 1 = \infty. \end{aligned}$$

Taking expectations gives a contradiction. Hence at least one state is recurrent, and irreducibility makes every state recurrent. $\square$

4.6 Stationary and Limiting Distributions

4.6.1 Stationarity

Definition 4.6.1 (Stationary distribution). A probability vector $\pi = (\pi_j)_{j \in S}$ is a stationary distribution for a Markov chain with transition matrix P if

$$\pi = \pi P, \quad \sum_{j \in S} \pi_j = 1, \quad \pi_j \geq 0.$$

Equivalently,

$$\pi_j = \sum_{i \in S} \pi_i p_{ij} \quad \text{for every } j.$$

If X_0 has distribution π , then X_n has distribution π for every n because

$$\pi P^n = \pi.$$

For a finite chain, the column vector $\pi^\top$ is an eigenvector of $P^\top$ with eigenvalue 1.

Definition 4.6.2 (Limiting distribution). A probability vector π is a limiting distribution if

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j$$

for every pair of states i, j . In this case the limiting law is independent of the initial state.

Whenever these limits exist in a finite chain, Chapman–Kolmogorov implies that the limiting vector is stationary. The converse is not automatic: a stationary distribution can exist even when P^n does not converge.

4.6.2 Two obstructions to convergence

Reducibility. If the state space has more than one closed communicating class, the long-run behavior can depend on the starting class. Thus irreducibility is a natural requirement when seeking a limit independent of the initial state.

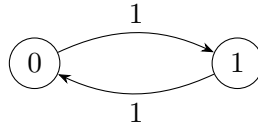
Periodicity. Consider the two-state chain

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$P^{2n} = I, \quad P^{2n+1} = P.$$

The distribution alternates between the two states and does not converge, although the stationary distribution $(1/2, 1/2)$ exists.



4.6.3 Period and aperiodicity

Definition 4.6.3 (Period). For a recurrent state i , let

$$D_i := \{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

The period of i is

$$d(i) := \gcd D_i.$$

A state is *aperiodic* if $d(i) = 1$.

If $p_{ii} > 0$, then $1 \in D_i$, so i is aperiodic. More generally, if two possible return times are relatively prime, the state is aperiodic. For example, possible returns in both 3 and 4 steps imply period 1.

Theorem 4.6.4 (Periodicity is a class property). *If $i \leftrightarrow j$, then*

$$d(i) = d(j).$$

Thus every state in an irreducible chain has the same period.

Proof. Choose r, s with $p_{ij}^{(r)} > 0$ and $p_{ji}^{(s)} > 0$. Then $r + s \in D_i$ and $r + s \in D_j$. For any $n \in D_i$,

$$p_{jj}^{(s+n+r)} \geq p_{ji}^{(s)} p_{ii}^{(n)} p_{ij}^{(r)} > 0,$$

so $s + n + r \in D_j$. Since $d(j)$ divides both $s + n + r$ and $s + r$, it divides n . Hence $d(j)$ divides every element of D_i , so $d(j) \mid d(i)$. Reversing the roles of i and j gives $d(i) \mid d(j)$. $\square$

4.6.4 Positive and null recurrence

For a recurrent state i , the return time T_i^+ is finite almost surely, but its expectation may still be infinite.

Definition 4.6.5 (Positive and null recurrence). A recurrent state i is

- *positive recurrent* if

$$m_i := \mathbb{E}_i T_i^+ < \infty;$$

- *null recurrent* if

$$\mathbb{E}_i T_i^+ = \infty.$$

Positive recurrence is a class property. Moreover, every recurrent state in a finite-state chain is positive recurrent. In particular, every finite irreducible chain is positive recurrent.

The importance of positive recurrence can be understood through long-run frequency. If state i occurs with limiting frequency π_i , then the average time between visits to i should be approximately $1/\pi_i$. An infinite mean return time would force $\pi_i = 0$; an irreducible stationary probability distribution cannot assign zero mass to every state.

4.6.5 The convergence theorem

Definition 4.6.6 (Ergodic state and ergodic chain). A positive recurrent, aperiodic state is called ergodic. An irreducible chain is called ergodic if its states are positive recurrent and aperiodic.

Theorem 4.6.7 (Limiting probabilities for an ergodic chain). *Let $\{X_n\}$ be an irreducible, positive recurrent, aperiodic Markov chain on a finite or countable state space. Then there is a unique stationary distribution π , with $\pi_j > 0$ for every j , and*

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j \quad \text{for all } i, j.$$

The stationary probabilities also satisfy

$$\pi_j = \frac{1}{m_j}, \quad m_j = \mathbb{E}_j T_j^+.$$

The theorem explains the numerical behavior of the phone model: under the stated conditions, every row of P^n converges to the same vector π .

Remark 4.6.8 (Why the stationary equation appears). Suppose $p_{ij}^{(n)} \rightarrow \pi_j$ independently of i . From

$$\mathbb{P}(X_{n+1} = j) = \sum_i \mathbb{P}(X_n = i) p_{ij},$$

passing to the limit gives

$$\pi_j = \sum_i \pi_i p_{ij}.$$

Thus any limiting distribution must satisfy $\pi = \pi P$.

4.6.6 Examples of stationary distributions

Example 4.6.9 (The two-state phone model). For

$$P = \begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix},$$

write $\pi = (\pi_0, \pi_1)$. The equations $\pi = \pi P$ and $\pi_0 + \pi_1 = 1$ give

$$\pi_0 = \frac{\beta}{\beta + 1 - \alpha}, \quad \pi_1 = \frac{1 - \alpha}{\beta + 1 - \alpha}.$$

For $\alpha = 0.7$ and $\beta = 0.4$,

$$\pi = \left(\frac{4}{7}, \frac{3}{7} \right) \approx (0.5714, 0.4286),$$

which agrees with the rows of P^8 computed earlier.

Example 4.6.10 (A simple model of class mobility). Let states 0, 1, 2 represent the upper, middle, and lower classes, respectively, and take

$$P = \begin{pmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{pmatrix}.$$

Solving $\pi = \pi P$ and $\pi_0 + \pi_1 + \pi_2 = 1$ gives approximately

$$\pi \approx (0.0624, 0.6234, 0.3142).$$

Thus, in the long run, about 6.2% of the population is in state 0, 62.3% in state 1, and 31.4% in state 2, according to this model.

4.6.7 Long-run averages and rewards

Theorem 4.6.11 (Markov-chain ergodic average). *Let $\{X_n\}$ be an irreducible positive recurrent Markov chain with stationary distribution π , and let r be a bounded function on the state space. Then*

$$\frac{1}{N} \sum_{n=1}^N r(X_n) \longrightarrow \sum_{j \in S} r(j) \pi_j$$

almost surely as $N \rightarrow \infty$.

If a reward $r(j)$ is earned whenever the chain visits state j , then the theorem says that the long-run average reward per unit time is

$$\sum_j r(j) \pi_j.$$

Taking $r = \mathbf{1}_{\{j\}}$ shows that the long-run fraction of time spent in state j is π_j .

4.7 Application: The Gambler's Ruin Problem

4.7.1 The model and state classification

At each game, a gambler wins one dollar with probability p and loses one dollar with probability

$$q = 1 - p.$$

The gambler starts with i dollars, where $0 < i < N$, and stops upon reaching either 0 or N . Let X_n denote the fortune after n games. The state space is

$$\{0, 1, \dots, N\},$$

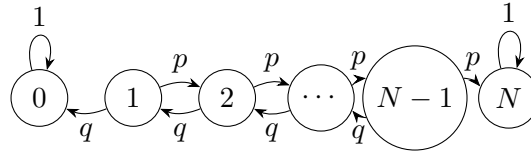
and the transition probabilities are

$$p_{j,j+1} = p, \quad p_{j,j-1} = q, \quad 1 \leq j \leq N-1,$$

together with

$$p_{00} = p_{NN} = 1.$$

Thus 0 and N are absorbing recurrent states, while $1, \dots, N-1$ are transient. Since a finite chain cannot remain forever among transient states, the gambler eventually reaches one of the two boundaries with probability one.



4.7.2 Probability of Reaching N before 0

Define

$$u_i := \mathbb{P}_i(\text{the chain reaches } N \text{ before } 0).$$

Conditioning on the first step gives

$$u_i = pu_{i+1} + qu_{i-1}, \quad 1 \leq i \leq N-1,$$

with boundary conditions

$$u_0 = 0, \quad u_N = 1.$$

For $p \neq q$, rearrange the recurrence as

$$p(u_{i+1} - u_i) = q(u_i - u_{i-1}).$$

If $d_i = u_i - u_{i-1}$, then

$$d_{i+1} = \frac{q}{p}d_i,$$

so

$$d_i = \left(\frac{q}{p}\right)^{i-1} d_1.$$

Summing the increments,

$$u_i = d_1 \sum_{k=0}^{i-1} \left(\frac{q}{p}\right)^k.$$

The condition $u_N = 1$ determines d_1 and gives

$$u_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}, \quad p \neq q.$$

If $p = q = 1/2$, the recurrence says that u_i is linear in i . The boundary conditions therefore give

$$u_i = \frac{i}{N}, \quad p = q = \frac{1}{2}.$$

The two formulas show the effect of drift. When $p > q$, the probability of reaching the upper boundary is larger than in the fair game; when $p < q$, it is smaller.

4.7.3 Letting the upper boundary tend to infinity

To display the dependence on the upper boundary, write

$$u_i^{(N)} := \mathbb{P}_i(\tau_N < \tau_0),$$

where

$$\tau_a := \inf\{n \geq 0 : X_n = a\}.$$

For fixed i and $N > i$, the finite-boundary formula is

$$u_i^{(N)} = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N}, & p \neq q, \\ \frac{i}{N}, & p = q = \frac{1}{2}. \end{cases}$$

Letting $N \rightarrow \infty$ while i remains fixed gives

$$\lim_{N \rightarrow \infty} u_i^{(N)} = \begin{cases} 1 - (q/p)^i, & p > \frac{1}{2}, \\ 0, & p \leq \frac{1}{2}. \end{cases}$$

Thus an upward drift leaves a positive probability of reaching arbitrarily high levels before ruin, whereas a fair or downward-drifting walk is eventually ruined with probability one.

4.8 Mean Time Spent in Transient States

4.8.1 The transient submatrix and the fundamental matrix

Suppose that a finite-state Markov chain is numbered so that

$$T = \{1, 2, \dots, t\}$$

is the set of all transient states. Let

$$Q = (p_{ij})_{i,j \in T}$$

be the submatrix of transition probabilities between states in T . The matrix Q is *sub-stochastic*: its entries are nonnegative and each row sum is at most 1. Probability mass missing from a row corresponds to a one-step transition from T to its complement.

For $i, j \in T$, define the expected occupation time

$$g_{ij} := \mathbb{E}_i \left[\sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=j\}} \right].$$

Thus g_{ij} is the expected total number of visits to j over the entire trajectory, including the initial time when $i = j$. Once the chain leaves the set of all transient states, it cannot return to it. Let

$$G_T = (g_{ij})_{i,j \in T}.$$

Theorem 4.8.1 (Fundamental matrix of the transient states). *For a finite transient set T ,*

$$G_T = I + QG_T = (I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n.$$

In particular, the entry $(G_T)_{ij}$ is the expected total time spent in state j when the chain starts from i .

Proof. Condition on the first step. Starting from i , one visit to j occurs immediately when $i = j$. If the first step is to a transient state k , the expected number of subsequent visits to j is g_{kj} . If the first step leaves T , no later visit to j is possible when T is the complete transient class under consideration. Hence

$$g_{ij} = \delta_{ij} + \sum_{k \in T} p_{ik} g_{kj}.$$

In matrix form,

$$G_T = I + QG_T.$$

Because every state in T is transient, the series $\sum_{n \geq 0} Q^n$ converges, and solving the matrix equation gives

$$G_T = (I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n.$$

□

Remark 4.8.2. The (i, j) entry of Q^n is the probability that the chain is in j after n steps while all intermediate evolution remains inside the transient subsystem. Summing these probabilities over n gives the expected occupation time.

4.8.2 Example: occupation times in gambler's ruin

Consider gambler's ruin with

$$p = 0.4, \quad q = 0.6, \quad N = 7,$$

and suppose the gambler starts with 3 dollars. The transient states are

$$T = \{1, 2, 3, 4, 5, 6\},$$

and the transient submatrix is

$$Q = \begin{pmatrix} 0 & 0.4 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 0.6 & 0 \end{pmatrix}.$$

The third row of the fundamental matrix $G_T = (I - Q)^{-1}$ is

$$(g_{3,1}, g_{3,2}, g_{3,3}, g_{3,4}, g_{3,5}, g_{3,6}) = \frac{1}{2059}(2925, 4875, 6175, 3610, 1900, 760).$$

Consequently,

$$g_{3,5} = \frac{1900}{2059} \approx 0.9228, \quad g_{3,2} = \frac{4875}{2059} \approx 2.3677.$$

These are the expected total numbers of time steps during which the gambler has 5 dollars and 2 dollars, respectively, before absorption.

4.8.3 Hitting probabilities from occupation times

For transient states $i, j \in T$, retain the notation

$$f_{ij} = \mathbb{P}_i(T_j^+ < \infty), \quad T_j^+ = \inf\{n \geq 1 : X_n = j\}.$$

When $i \neq j$, this is the probability of ever visiting j . When $i = j$, it is the probability of returning to j after time 0.

Proposition 4.8.3 (Hitting probability from the fundamental matrix). *For transient states i, j ,*

$$f_{ij} = \frac{g_{ij} - \delta_{ij}}{g_{jj}}.$$

In particular,

$$f_{ij} = \frac{g_{ij}}{g_{jj}} \quad (i \neq j), \quad f_{ii} = 1 - \frac{1}{g_{ii}}.$$

Proof. Condition on whether the chain reaches j after leaving its starting point. On the hitting event, the strong Markov property shows that the expected number of visits from the hitting time onward is g_{jj} . The initial state contributes an additional δ_{ij} . On the event of no positive-time hit, the only possible visit is the initial one when $i = j$. Therefore

$$g_{ij} = (\delta_{ij} + g_{jj})f_{ij} + \delta_{ij}(1 - f_{ij}) = \delta_{ij} + g_{jj}f_{ij},$$

which gives the stated formula. □

4.9 The Galton–Watson Branching Process

The branching process introduced earlier is now studied in more detail. To distinguish the population size from a generic Markov chain X_n , write Z_n for the population in generation n .

4.9.1 Definition and elementary state classification

Let

$$Z_0 = 1, \quad Z_n = \sum_{i=1}^{Z_{n-1}} \xi_{n,i}, \quad n \geq 1,$$

where an empty sum is interpreted as 0. The random variable $\xi_{n,i}$ is the number of offspring of the i th individual in generation $n - 1$. Assume that the family $\{\xi_{n,i}\}$ is iid and write

$$p_k := \mathbb{P}(\xi_{n,i} = k), \quad k = 0, 1, 2, \dots$$

We mainly focus on the nondegenerate case in which

$$p_k < 1 \quad \text{for every } k, \quad p_0 > 0.$$

Proposition 4.9.1 (Markov and state properties). *Under the preceding assumptions:*

- (i) $\{Z_n\}_{n \geq 0}$ is a Markov chain on $\mathbb{N}_0$;
- (ii) state 0 is absorbing and hence recurrent;
- (iii) every state $i \geq 1$ is transient;
- (iv) every finite set of positive states is visited only finitely many times almost surely.

Proof. Conditional on $Z_{n-1} = i$, the next generation is a sum of i iid offspring variables, so its law depends only on i . Hence the process is Markov. If $Z_n = 0$, then $Z_m = 0$ for all $m \geq n$, so 0 is absorbing. From a positive state i ,

$$\mathbb{P}_i(Z_1 = 0) = p_0^i > 0,$$

whereas state 0 cannot reach any positive state. A recurrent state cannot access a state from which it cannot be reached back, so every positive state is transient. A transient state is visited only finitely often almost surely; the same is therefore true for a finite union of positive states. $\square$

4.9.2 Mean and variance of the population

Assume that the offspring variable ξ has finite mean and variance

$$\mu := \mathbb{E}\xi = \sum_{k=0}^{\infty} kp_k, \quad \sigma^2 := \text{Var}(\xi) = \sum_{k=0}^{\infty} (k - \mu)^2 p_k.$$

Conditional on Z_{n-1} ,

$$\mathbb{E}(Z_n \mid Z_{n-1}) = \mu Z_{n-1}, \quad \text{Var}(Z_n \mid Z_{n-1}) = \sigma^2 Z_{n-1}.$$

Proposition 4.9.2 (Population moments). *For a Galton–Watson process started from one ancestor,*

$$\boxed{\mathbb{E}Z_n = \mu^n.}$$

Moreover,

$$\boxed{\text{Var}(Z_n) = \begin{cases} \sigma^2 \mu^{n-1} \frac{1 - \mu^n}{1 - \mu}, & \mu \neq 1, \\ n\sigma^2, & \mu = 1. \end{cases}}$$

Proof. The law of total expectation gives

$$\mathbb{E}Z_n = \mathbb{E}[\mathbb{E}(Z_n \mid Z_{n-1})] = \mu\mathbb{E}Z_{n-1}.$$

Since $Z_0 = 1$, iteration yields $\mathbb{E}Z_n = \mu^n$.

For the variance, the law of total variance gives

$$\begin{aligned} \text{Var}(Z_n) &= \mathbb{E}[\text{Var}(Z_n \mid Z_{n-1})] + \text{Var}(\mathbb{E}(Z_n \mid Z_{n-1})) \\ &= \sigma^2\mathbb{E}Z_{n-1} + \mu^2\text{Var}(Z_{n-1}) \\ &= \sigma^2\mu^{n-1} + \mu^2\text{Var}(Z_{n-1}). \end{aligned}$$

Because $\text{Var}(Z_0) = 0$, iteration gives

$$\text{Var}(Z_n) = \sigma^2(\mu^{n-1} + \mu^n + \cdots + \mu^{2n-2}),$$

which is the stated formula. □

4.9.3 The extinction event

Define

$$\mathcal{E} := \{\text{the population eventually becomes extinct}\} = \bigcup_{n \geq 0} \{Z_n = 0\}.$$

Since state 0 is absorbing, the events $\{Z_n = 0\}$ increase with n . Therefore

$$\eta := \mathbb{P}(\mathcal{E}) = \lim_{n \rightarrow \infty} \mathbb{P}(Z_n = 0).$$

When $\mu < 1$, Markov's inequality gives

$$\mathbb{P}(Z_n \geq 1) \leq \mathbb{E}Z_n = \mu^n \longrightarrow 0,$$

and hence

$$\boxed{\eta = 1 \quad (\mu < 1).}$$

For the critical and supercritical cases, probability generating functions provide the natural tool.

4.10 Probability Generating Functions

4.10.1 Definition and basic identities

Definition 4.10.1 (Probability generating function). If X is a nonnegative integer-valued random variable with PMF $p_X(k) = \mathbb{P}(X = k)$, its probability generating function, or PGF, is

$$\boxed{G_X(s) := \mathbb{E}[s^X] = \sum_{k=0}^{\infty} p_X(k)s^k, \quad 0 \leq s \leq 1.}$$

The coefficients can be recovered from derivatives at the origin:

$$p_X(k) = \frac{G_X^{(k)}(0)}{k!}, \quad k = 0, 1, 2, \dots$$

The relation with the moment-generating function from Chapter 2 is

$$M_X(t) = G_X(e^t)$$

whenever both sides are finite.

Proposition 4.10.2 (Factorial moments from a PGF). *Whenever the corresponding moments are finite,*

$$\mathbb{E}X = G'_X(1^-)$$

and, for $r \geq 1$,

$$\mathbb{E}[X(X-1)\cdots(X-r+1)] = G_X^{(r)}(1^-).$$

Consequently,

$$\text{Var}(X) = G''_X(1^-) + G'_X(1^-) - (G'_X(1^-))^2.$$

Proof. Differentiate the power series term by term for $0 \leq s < 1$ and then let $s \uparrow 1$. The variance identity follows from

$$X^2 = X(X-1) + X.$$

□

4.10.2 Sums of independent random variables

Proposition 4.10.3 (PGF of an independent sum). *If X and Y are independent nonnegative integer-valued random variables, then*

$$G_{X+Y}(s) = G_X(s)G_Y(s).$$

More generally, for independent $X_1, \dots, X_n$,

$$G_{X_1+\dots+X_n}(s) = \prod_{i=1}^n G_{X_i}(s).$$

Proof. Independence gives

$$G_{X+Y}(s) = \mathbb{E}[s^{X+Y}] = \mathbb{E}[s^X s^Y] = \mathbb{E}[s^X] \mathbb{E}[s^Y].$$

□

Example 4.10.4 (Sum of independent Poisson variables). If

$$X \sim \text{Poisson}(\lambda), \quad Y \sim \text{Poisson}(\nu), \quad X \perp\!\!\!\perp Y,$$

then

$$G_X(s) = e^{\lambda(s-1)}, \quad G_Y(s) = e^{\nu(s-1)}.$$

Thus

$$G_{X+Y}(s) = e^{(\lambda+\nu)(s-1)},$$

so

$$X + Y \sim \text{Poisson}(\lambda + \nu).$$

4.10.3 A random number of summands

Proposition 4.10.5 (Compound-sum PGF). *Let $Y_1, Y_2, \dots$ be iid nonnegative integer-valued random variables with common PGF G_Y . Let N be a nonnegative integer-valued random variable, independent of the entire sequence, with PGF G_N . For*

$$S_N := \sum_{i=1}^N Y_i,$$

where the empty sum is 0,

$$G_{S_N}(s) = G_N(G_Y(s)).$$

Proof. Conditioning on N gives

$$\begin{aligned} G_{S_N}(s) &= \mathbb{E}[\mathbb{E}(s^{S_N} \mid N)] \\ &= \mathbb{E}[(G_Y(s))^N] \\ &= G_N(G_Y(s)). \end{aligned}$$

□

Example 4.10.6 (Poisson thinning). Suppose

$$N \sim \text{Poisson}(\lambda),$$

and each of the N objects is retained independently with probability p . Let $Y_i \sim \text{Bernoulli}(p)$ be the indicator that object i is retained, and let

$$K = \sum_{i=1}^N Y_i.$$

Since

$$G_Y(s) = 1 - p + ps,$$

we obtain

$$\begin{aligned} G_K(s) &= G_N(G_Y(s)) \\ &= \exp\{\lambda[(1 - p + ps) - 1]\} \\ &= e^{\lambda p(s-1)}. \end{aligned}$$

Therefore

$$K \sim \text{Poisson}(\lambda p).$$

This is the Poisson thinning property.

4.11 Extinction via Probability Generating Functions

Return to the Galton–Watson process and let

$$G(s) := \mathbb{E}[s^\xi] = \sum_{k=0}^{\infty} p_k s^k$$

be the offspring PGF. For the generation size, write

$$G_n(s) := \mathbb{E}[s^{Z_n}], \quad G_0(s) = s.$$

Proposition 4.11.1 (Iteration of the offspring PGF). *For $n \geq 1$,*

$$\boxed{G_n(s) = G_{n-1}(G(s)) = G^{\circ n}(s),}$$

where $G^{\circ n}$ denotes n -fold composition and $G^{\circ 0}(s) = s$. More generally,

$$G_n = G_m \circ G_{n-m}, \quad 0 \leq m \leq n.$$

Proof. Conditional on Z_{n-1} , the next generation is a sum of Z_{n-1} iid offspring variables. The compound-sum formula therefore gives

$$G_n(s) = G_{n-1}(G(s)).$$

Starting from $G_0(s) = s$ and iterating proves the result. □

Set

$$\eta_n := \mathbb{P}(Z_n = 0) = G_n(0).$$

Then $\eta_n \uparrow \eta$ and

$$\eta_{n+1} = G(\eta_n), \quad \eta_0 = 0.$$

Theorem 4.11.2 (Fixed-point characterization of extinction). *The extinction probability η is the smallest solution in $[0, 1]$ of*

$$\boxed{s = G(s).}$$

Starting from m independent initial ancestors, the extinction probability is η^m .

Proof. By continuity of G on $[0, 1]$,

$$\eta = \lim_{n \rightarrow \infty} \eta_{n+1} = \lim_{n \rightarrow \infty} G(\eta_n) = G(\eta),$$

so η is a fixed point. If $r \in [0, 1]$ is any fixed point, then $\eta_0 = 0 \leq r$ and monotonicity of G gives inductively

$$\eta_n \leq r \quad \text{for every } n.$$

Hence $\eta \leq r$, proving minimality. With m initial ancestors, the m descendant trees are independent, so all of them become extinct with probability η^m . □

Theorem 4.11.3 (Extinction classification). *Let*

$$\mu = G'(1^-) = \mathbb{E}\xi, \quad \sigma^2 = \text{Var}(\xi).$$

For a process started from one ancestor,

<i>offspring law</i>	<i>extinction probability</i>
$\mu < 1$	$\eta = 1$
$\mu > 1$	$\eta < 1$
$\mu = 1, \sigma^2 > 0$	$\eta = 1$
$\mu = 1, \sigma^2 = 0$	$\eta = 0$

In the last case, $\xi = 1$ almost surely and the population remains equal to 1 forever.

Proof. The subcritical case $\mu < 1$ was already proved from $\mathbb{P}(Z_n \geq 1) \leq \mu^n$.

The PGF is increasing and convex on $[0, 1]$, with

$$G(1) = 1, \quad G'(1^-) = \mu.$$

If $\mu > 1$, then for $s < 1$ sufficiently close to 1,

$$G(s) - s < 0,$$

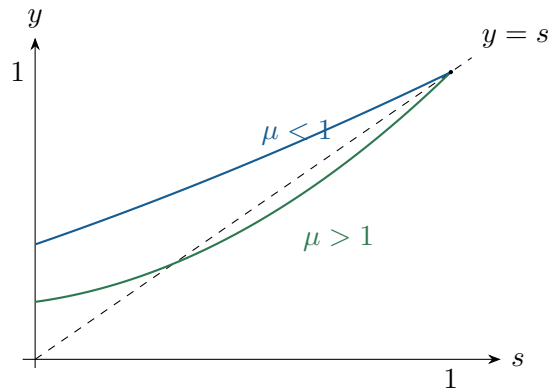
whereas $G(0) - 0 = p_0 \geq 0$. Hence there is a fixed point below 1; if $p_0 = 0$, that fixed point is 0. The smallest fixed point therefore satisfies $\eta < 1$.

If $\mu = 1$ and $\sigma^2 > 0$, the offspring law is nondegenerate and G is strictly convex. Its tangent line at $s = 1$ is the line $y = s$, so strict convexity gives

$$G(s) > s, \quad 0 \leq s < 1.$$

Thus the only fixed point in $[0, 1]$ is 1, and $\eta = 1$.

Finally, if $\mu = 1$ and $\sigma^2 = 0$, then $\xi = 1$ almost surely. Hence $G(s) = s$ and $Z_n = 1$ for every n , so extinction never occurs and $\eta = 0$. $\square$



The diagram illustrates why a supercritical PGF has a fixed point below 1, while a subcritical nondegenerate PGF meets the diagonal only at 1.

4.12 Return-Time Generating Functions and Null Recurrence

The earlier recurrence test used the series $\sum_n p_{00}^{(n)}$. Generating functions give a sharper description of the first return time and show that the one-dimensional symmetric walk is null recurrent.

Let

$$S_n = \sum_{i=1}^n Y_i, \quad S_0 = 0,$$

where the increments are iid with

$$\mathbb{P}(Y_i = 1) = p, \quad \mathbb{P}(Y_i = -1) = q = 1 - p.$$

Define the first return time

$$T_0^+ := \inf\{n \geq 1 : S_n = 0\}.$$

In the transient case, $\mathbb{P}(T_0^+ = \infty) > 0$; thus T_0^+ is naturally regarded as an extended-valued, or defective, return time.

For $n \geq 0$, set

$$p_{00}^{(n)} := \mathbb{P}(S_n = 0),$$

and, for $n \geq 1$, set

$$f_{00}^{(n)} := \mathbb{P}(T_0^+ = n).$$

Introduce the generating functions

$$P_0(s) := \sum_{n=0}^{\infty} p_{00}^{(n)} s^n, \quad F_0(s) := \sum_{n=1}^{\infty} f_{00}^{(n)} s^n, \quad 0 \leq s < 1.$$

The function P_0 is not generally a PGF of a random variable because its coefficients need not sum to 1. The function F_0 is a defective PGF when the eventual return probability is less than 1.

Theorem 4.12.1 (Return and first-return generating functions). *For the one-dimensional nearest-neighbor walk,*

$$P_0(s) = 1 + P_0(s)F_0(s).$$

Moreover,

$$P_0(s) = \frac{1}{\sqrt{1 - 4pqs^2}}, \quad F_0(s) = 1 - \sqrt{1 - 4pqs^2}.$$

Proof. For $n \geq 1$, decompose a return at time n according to the first return time k :

$$p_{00}^{(n)} = \sum_{k=1}^n f_{00}^{(k)} p_{00}^{(n-k)}.$$

Multiplying by s^n and summing over $n \geq 1$ gives

$$P_0(s) - 1 = F_0(s)P_0(s).$$

A return is possible only after an even number of steps. For $m \geq 0$,

$$p_{00}^{(2m)} = \binom{2m}{m} (pq)^m, \quad p_{00}^{(2m+1)} = 0.$$

Using the central-binomial power series

$$\sum_{m=0}^{\infty} \binom{2m}{m} x^m = \frac{1}{\sqrt{1-4x}},$$

we obtain

$$P_0(s) = \sum_{m=0}^{\infty} \binom{2m}{m} (pqs^2)^m = \frac{1}{\sqrt{1-4pqs^2}}.$$

Solving the renewal identity for F_0 gives the second formula. □

Corollary 4.12.2 (Recurrence and null recurrence). *The eventual return probability is*

$$f_{00} := \mathbb{P}(T_0^+ < \infty) = F_0(1^-) = 1 - \sqrt{1-4pq} = 1 - |p - q|.$$

Consequently, the walk is recurrent exactly when $p = q = 1/2$. In the symmetric case,

$$\mathbb{E}_0 T_0^+ = F'_0(1^-) = \infty,$$

so the one-dimensional symmetric random walk is null recurrent.

Proof. Since $p + q = 1$,

$$\sqrt{1-4pq} = |p - q|.$$

Thus $f_{00} = 1$ exactly when $p = q = 1/2$. Also,

$$F'_0(s) = \frac{4pqs}{\sqrt{1-4pqs^2}}.$$

When $p = q = 1/2$, this becomes

$$F'_0(s) = \frac{s}{\sqrt{1-s^2}},$$

which diverges as $s \uparrow 1$. □

Chapter 4 Summary of Key Formulas

Markov property and one-step transitions

For a time-homogeneous Markov chain,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) = p_{ij}.$$

For every state i ,

$$p_{ij} \geq 0, \quad \sum_j p_{ij} = 1.$$

For a random walk $X_n = X_0 + \sum_{k=1}^n \xi_k$ with increment PMF q ,

$$p_{ij} = q(j - i).$$

Multi-step transitions and distributions

$$p_{ij}^{(n)} = \mathbb{P}(X_{m+n} = j \mid X_m = i), \quad p_{ij}^{(0)} = \mathbf{1}_{\{i=j\}}.$$

The Chapman–Kolmogorov equations are

$$p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)}.$$

For a finite chain,

$$p_{ij}^{(n)} = (P^n)_{ij}.$$

If μ_n is the row vector giving the law of X_n , then

$$\mu_{n+1} = \mu_n P, \quad \mu_n = \mu_0 P^n.$$

Communication, recurrence, and occupation counts

$$i \rightarrow j \iff p_{ij}^{(n)} > 0 \text{ for some } n \geq 0, \quad i \leftrightarrow j \iff i \rightarrow j \text{ and } j \rightarrow i.$$

A chain is irreducible if all states communicate. With

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}, \quad f_{ii} = \mathbb{P}_i(T_i^+ < \infty),$$

state i is recurrent when $f_{ii} = 1$ and transient when $f_{ii} < 1$. Also,

$$\mathbb{E}_i \left[\sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=i\}} \right] = \sum_{n=0}^{\infty} p_{ii}^{(n)},$$

so

$$i \text{ recurrent} \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty.$$

Recurrence, transience, and periodicity are class properties.

Random-walk recurrence

For the nearest-neighbor walk on $\mathbb{Z}$ with right-step probability p and $q = 1 - p$,

$$p_{00}^{(2n)} = \binom{2n}{n} (pq)^n \sim \frac{[4pq]^n}{\sqrt{\pi n}}.$$

Hence the walk is recurrent exactly when $p = q = 1/2$. For simple symmetric random walk on $\mathbb{Z}^d$,

$$d = 1, 2 : \text{ recurrent, } \quad d \geq 3 : \text{ transient.}$$

Every finite irreducible Markov chain is recurrent and positive recurrent.

Stationarity, period, and limiting probabilities

A stationary distribution satisfies

$$\pi = \pi P, \quad \sum_j \pi_j = 1, \quad \pi_j \geq 0.$$

The period of a recurrent state is

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

For an irreducible, positive recurrent, aperiodic chain,

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j, \quad \pi_j = \frac{1}{\mathbb{E}_j T_j^+}.$$

For bounded r ,

$$\frac{1}{N} \sum_{n=1}^N r(X_n) \longrightarrow \sum_j r(j) \pi_j \quad \text{almost surely.}$$

Gambler's ruin

Let $q = 1 - p$ and

$$u_i^{(N)} = \mathbb{P}_i(\tau_N < \tau_0).$$

Then

$$u_i^{(N)} = p u_{i+1}^{(N)} + q u_{i-1}^{(N)}, \quad u_0^{(N)} = 0, \quad u_N^{(N)} = 1,$$

and therefore

$$u_i^{(N)} = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N}, & p \neq q, \\ \frac{i}{N}, & p = q = 1/2. \end{cases}$$

For fixed i ,

$$\lim_{N \rightarrow \infty} u_i^{(N)} = \begin{cases} 1 - (q/p)^i, & p > 1/2, \\ 0, & p \leq 1/2. \end{cases}$$

Fundamental matrix of transient states

Let T be the finite set of transient states and let

$$Q = (p_{ij})_{i,j \in T}.$$

If

$$g_{ij} = \mathbb{E}_i \left[\sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=j\}} \right], \quad G_T = (g_{ij})_{i,j \in T},$$

then

$$G_T = (I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n.$$

For transient states i, j ,

$$f_{ij} = \frac{g_{ij} - \delta_{ij}}{g_{jj}}.$$

Probability generating functions

For a nonnegative integer-valued X ,

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k, \quad \mathbb{P}(X = k) = \frac{G_X^{(k)}(0)}{k!}.$$

When the moments are finite,

$$\mathbb{E}X = G'_X(1^-), \quad \text{Var}(X) = G''_X(1^-) + G'_X(1^-) - (G'_X(1^-))^2.$$

For independent X, Y ,

$$G_{X+Y} = G_X G_Y.$$

If $S_N = \sum_{i=1}^N Y_i$, where N is independent of iid Y_i , then

$$G_{S_N}(s) = G_N(G_Y(s)).$$

In particular, Poisson thinning gives

$$N \sim \text{Poisson}(\lambda), \quad Y_i \sim \text{Bernoulli}(p) \implies \sum_{i=1}^N Y_i \sim \text{Poisson}(\lambda p).$$

Galton–Watson moments and extinction

For

$$Z_n = \sum_{i=1}^{Z_{n-1}} \xi_{n,i}, \quad Z_0 = 1, \quad \mu = \mathbb{E}\xi, \quad \sigma^2 = \text{Var}(\xi),$$

we have

$$\mathbb{E}Z_n = \mu^n,$$

and

$$\text{Var}(Z_n) = \begin{cases} \sigma^2 \mu^{n-1} \frac{1 - \mu^n}{1 - \mu}, & \mu \neq 1, \\ n\sigma^2, & \mu = 1. \end{cases}$$

If G is the offspring PGF and η is the extinction probability, then

$$G_n = G^{on}, \quad \eta = \lim_{n \rightarrow \infty} G_n(0), \quad \boxed{\eta \text{ is the smallest solution of } s = G(s).}$$

The classification is

$$\boxed{\mu < 1 \Rightarrow \eta = 1, \quad \mu > 1 \Rightarrow \eta < 1, \quad \mu = 1, \sigma^2 > 0 \Rightarrow \eta = 1, \quad \mu = 1, \sigma^2 = 0 \Rightarrow \eta = 0.}$$

First-return generating functions

For one-dimensional nearest-neighbor random walk, let

$$P_0(s) = \sum_{n=0}^{\infty} p_{00}^{(n)} s^n, \quad F_0(s) = \sum_{n=1}^{\infty} \mathbb{P}_0(T_0^+ = n) s^n.$$

The first-return decomposition gives

$$P_0(s) = 1 + P_0(s)F_0(s).$$

Consequently,

$$\boxed{P_0(s) = \frac{1}{\sqrt{1 - 4pqs^2}}, \quad F_0(s) = 1 - \sqrt{1 - 4pqs^2}.}$$

Thus

$$\mathbb{P}_0(T_0^+ < \infty) = F_0(1^-) = 1 - |p - q|.$$

For $p = q = 1/2$,

$$\mathbb{P}_0(T_0^+ < \infty) = 1, \quad \mathbb{E}_0 T_0^+ = F_0'(1^-) = \infty,$$

so the symmetric walk is null recurrent.

Chapter 5

Poisson Processes

Scope of Chapter 5

This chapter studies continuous-time counting processes, with the homogeneous Poisson process as the central model. We begin by recalling the exponential distribution and its memoryless property, then give two equivalent definitions of a Poisson process. We develop interarrival and arrival times, thinning and superposition, exponential races, Poissonization of the coupon collector problem, conditional arrival-time laws and order statistics, and finally nonhomogeneous Poisson processes with time-dependent intensity.

5.1 The Exponential Distribution and Memorylessness

5.1.1 Basic formulas

The exponential law was introduced in Chapter 2. We record the formulas that will be used repeatedly in the study of Poisson processes.

Definition 5.1.1 (Exponential distribution). For a rate parameter $\lambda > 0$, a random variable X has the exponential distribution with rate λ , written

$$X \sim \text{Exp}(\lambda),$$

if its density is

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0, \infty)}(x).$$

Its c.d.f. and survival function are

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0, \end{cases} \quad \bar{F}_X(x) := \mathbb{P}(X > x) = e^{-\lambda x}, \quad x \geq 0.$$

For $u < \lambda$, the moment-generating function is

$$M_X(u) = \mathbb{E}[e^{uX}] = \frac{\lambda}{\lambda - u}.$$

Consequently,

$$\mathbb{E}X = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

5.1.2 The memoryless property

Definition 5.1.2 (Memoryless random variable). A nonnegative random variable X is called *memoryless* if

$$\mathbb{P}(X > s + t \mid X > t) = \mathbb{P}(X > s)$$

for all $s, t \geq 0$ for which the conditional probability is defined.

The interpretation is that the amount of time already spent waiting does not affect the distribution of the additional waiting time.

Proposition 5.1.3 (Exponential waiting times are memoryless). *If $X \sim \text{Exp}(\lambda)$, then X is memoryless.*

Proof. For $s, t \geq 0$,

$$\mathbb{P}(X > s + t \mid X > t) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > t)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = \mathbb{P}(X > s).$$

□

5.1.3 An infinitesimal characterization

The notation $o(h)$ means a quantity satisfying

$$\frac{o(h)}{h} \longrightarrow 0 \quad \text{as } h \downarrow 0.$$

Proposition 5.1.4 (Constant short-time rate characterizes the exponential law). *Let X be a nonnegative waiting time with $\mathbb{P}(X > 0) = 1$. Then*

$$X \sim \text{Exp}(\lambda)$$

if and only if, for every $t \geq 0$,

$$\mathbb{P}(t < X \leq t + h \mid X > t) = \lambda h + o(h) \quad \text{as } h \downarrow 0.$$

Proof. If $X \sim \text{Exp}(\lambda)$, memorylessness gives

$$\mathbb{P}(t < X \leq t + h \mid X > t) = 1 - \mathbb{P}(X > t + h \mid X > t) = 1 - e^{-\lambda h} = \lambda h + o(h).$$

Conversely, write $g(t) = \mathbb{P}(X > t)$. The assumed relation is equivalent to

$$\mathbb{P}(X > t + h \mid X > t) = 1 - \lambda h + o(h).$$

Since $\{X > t + h\} \subseteq \{X > t\}$,

$$g(t + h) = \mathbb{P}(X > t + h \mid X > t)g(t) = (1 - \lambda h + o(h))g(t).$$

Therefore

$$\frac{g(t + h) - g(t)}{h} = -\lambda g(t) + \frac{o(h)}{h}g(t),$$

and hence

$$g'(t) = -\lambda g(t), \quad g(0) = 1.$$

The solution is $g(t) = e^{-\lambda t}$, which is the survival function of $\text{Exp}(\lambda)$. □

Remark 5.1.5 (Hazard-rate interpretation). The conditional probability of an event during the next short interval of length h , given survival up to time t , is approximately λh . Thus the exponential distribution has a constant instantaneous event rate.

5.2 Counting Processes and the Poisson Process

5.2.1 Counting processes

Definition 5.2.1 (Counting process). A stochastic process

$$\{N(t) : t \geq 0\}$$

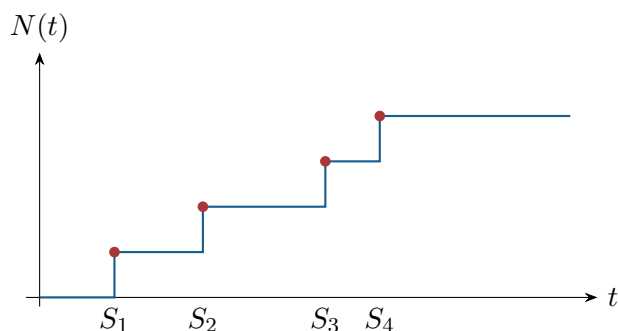
is called a *counting process* if $N(t)$ represents the total number of events that have occurred by time t .

For example, $N(t)$ may count the number of customers who have entered a store by time t , or the number of births recorded by time t .

A counting process has the following basic properties:

- $N(t) \geq 0$;
- $N(t)$ is integer-valued;
- $N(s) \leq N(t)$ whenever $0 \leq s < t$;
- $N(t) - N(s)$ is the number of events in the interval $(s, t]$.

A sample path is therefore a nondecreasing step function. A typical path has the form shown below.



Definition 5.2.2 (Independent increments). A counting process has *independent increments* if the numbers of events in disjoint time intervals are independent.

Definition 5.2.3 (Stationary increments). A counting process has *stationary increments* if the distribution of

$$N(t) - N(s)$$

depends only on the interval length $t - s$.

Disjoint intervals of the same length need not have identically distributed counts for a general counting process. Independent and stationary increments are additional structural assumptions.

5.2.2 First definition of a homogeneous Poisson process

Definition 5.2.4 (Poisson process: increment-distribution definition). A counting process $\{N(t) : t \geq 0\}$ is a *Poisson process with rate $\lambda > 0$* if

1. $N(0) = 0$;
2. the process has independent increments;
3. for all $s, t \geq 0$,

$$N(s+t) - N(s) \sim \text{Poisson}(\lambda t).$$

The third condition implies stationary increments. In particular,

$$N(t) \sim \text{Poisson}(\lambda t),$$

so

$$\mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots,$$

and

$$\mathbb{E}N(t) = \text{Var}(N(t)) = \lambda t.$$

Its moment-generating function is

$$\mathbb{E}[e^{uN(t)}] = \exp\{\lambda t(e^u - 1)\}.$$

5.2.3 Second definition: local behavior

Definition 5.2.5 (Poisson process: infinitesimal definition). A counting process $\{N(t) : t \geq 0\}$ is a Poisson process with rate $\lambda > 0$ if

1. $N(0) = 0$;
2. it has stationary and independent increments;
3. as $h \downarrow 0$,

$$\mathbb{P}(N(h) = 1) = \lambda h + o(h);$$

4. as $h \downarrow 0$,

$$\mathbb{P}(N(h) \geq 2) = o(h).$$

The last two conditions imply

$$\mathbb{P}(N(h) = 0) = 1 - \lambda h + o(h).$$

Heuristically, one event occurs with probability of order h , while two or more events occur with a probability negligible compared with h .

Theorem 5.2.6 (Equivalence of the two definitions). *The increment-distribution definition and the infinitesimal definition describe the same class of counting processes.*

Proof. Suppose first that

$$N(h) \sim \text{Poisson}(\lambda h).$$

Then

$$\mathbb{P}(N(h) = 1) = \lambda h e^{-\lambda h} = \lambda h + o(h),$$

and

$$\begin{aligned}\mathbb{P}(N(h) \geq 2) &= 1 - \mathbb{P}(N(h) = 0) - \mathbb{P}(N(h) = 1) \\ &= 1 - e^{-\lambda h} - \lambda h e^{-\lambda h} = o(h).\end{aligned}$$

Thus the first definition implies the second.

Conversely, assume the infinitesimal definition. For $u \leq 0$, let

$$g(t, u) = \mathbb{E}[e^{uN(t)}].$$

By stationary and independent increments,

$$g(t + h, u) = g(t, u)\mathbb{E}[e^{uN(h)}].$$

The small-time assumptions give

$$\begin{aligned}\mathbb{E}[e^{uN(h)}] &= \mathbb{P}(N(h) = 0) + e^u \mathbb{P}(N(h) = 1) + \sum_{k \geq 2} e^{uk} \mathbb{P}(N(h) = k) \\ &= 1 - \lambda h + \lambda h e^u + o(h) \\ &= 1 + \lambda h(e^u - 1) + o(h),\end{aligned}$$

where $u \leq 0$ ensures that the contribution from $N(h) \geq 2$ is bounded by $o(h)$. Hence

$$\frac{g(t + h, u) - g(t, u)}{h} = \lambda(e^u - 1)g(t, u) + o(1).$$

Letting $h \downarrow 0$ gives

$$\frac{\partial}{\partial t} g(t, u) = \lambda(e^u - 1)g(t, u), \quad g(0, u) = 1.$$

Therefore

$$g(t, u) = \exp\{\lambda t(e^u - 1)\},$$

which is the transform of a $\text{Poisson}(\lambda t)$ random variable. Thus $N(t) \sim \text{Poisson}(\lambda t)$, and stationary increments give

$$N(s + t) - N(s) \sim \text{Poisson}(\lambda t).$$

□

5.3 Interarrival Times and Arrival Times

5.3.1 Interarrival times

Let S_n denote the time of the n th event, with $S_0 = 0$. Define the interarrival times by

$$T_n = S_n - S_{n-1}, \quad n \geq 1.$$

Thus T_n is the elapsed time between the $(n - 1)$ st and n th events.

Theorem 5.3.1 (Interarrival-time representation). *For a Poisson process with rate λ , the variables*

$$T_1, T_2, \dots$$

are independent and identically distributed with

$$T_n \sim \text{Exp}(\lambda).$$

Proof. For the first waiting time,

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N(t) = 0) = e^{-\lambda t},$$

so $T_1 \sim \text{Exp}(\lambda)$.

After an arrival, future increments of the Poisson process are independent of all increments before that arrival and have the same distribution as increments starting at time 0. Equivalently, conditional on an event having occurred at time S_{n-1} , the probability of no further event during the next t units of time is

$$\mathbb{P}(N(S_{n-1} + t) - N(S_{n-1}) = 0) = e^{-\lambda t}.$$

Applying this argument successively shows that every T_n is $\text{Exp}(\lambda)$ and that the interarrival times are independent. $\square$

The counting and waiting-time descriptions are related by

$$S_n = T_1 + \dots + T_n$$

and

$$N(t) = \sum_{n=1}^{\infty} \mathbf{1}_{\{S_n \leq t\}}.$$

Also,

$$\mathbb{E}T_1 = \frac{1}{\lambda}, \quad \mathbb{E}N(t) = \lambda t = \frac{t}{\mathbb{E}T_1}.$$

5.3.2 The Waiting Time Until the n th Event

Definition 5.3.2 (Arrival time). The random variable

$$S_n = T_1 + \dots + T_n$$

is the *arrival time* of the n th event, or the waiting time until the n th event.

Proposition 5.3.3 (Gamma law for arrival times). *For $n \geq 1$,*

$$S_n \sim \text{Gamma}(n, \lambda).$$

More explicitly,

$$\mathbb{P}(S_n \leq t) = \mathbb{P}(N(t) \geq n) = 1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}, \quad t \geq 0,$$

and

$$f_{S_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t > 0.$$

Proof. The event $\{S_n \leq t\}$ is the same as $\{N(t) \geq n\}$. The c.d.f. follows from the Poisson distribution, and differentiation gives the density. Equivalently, S_n is a sum of n independent $\text{Exp}(\lambda)$ variables. $\square$

A local derivation of the same density is useful. For small $h > 0$,

$$\begin{aligned} \mathbb{P}(t < S_n \leq t + h) &= \mathbb{P}(N(t) = n - 1, N(t + h) - N(t) = 1) + o(h) \\ &= \mathbb{P}(N(t) = n - 1)(\lambda h + o(h)) + o(h) \\ &= \left[\lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} \right] h + o(h). \end{aligned}$$

5.4 Splitting, Thinning, and Superposition

5.4.1 Independent classification of events

Suppose that each event of a Poisson process is independently classified as type I with probability p and type II with probability $1 - p$. Let $N_1(t)$ and $N_2(t)$ be the numbers of type I and type II events by time t . Then

$$N(t) = N_1(t) + N_2(t).$$

Theorem 5.4.1 (Poisson splitting). *If N is a Poisson process with rate λ , then*

$$\{N_1(t) : t \geq 0\} \quad \text{and} \quad \{N_2(t) : t \geq 0\}$$

are independent Poisson processes with rates

$$\lambda p \quad \text{and} \quad \lambda(1 - p),$$

respectively.

Proof. The independent marking operation preserves stationary and independent increments. For a short interval of length h ,

$$\begin{aligned} \mathbb{P}(N_1(h) = 1) &= p\mathbb{P}(N(h) = 1) + O(\mathbb{P}(N(h) \geq 2)) \\ &= \lambda p h + o(h), \end{aligned}$$

while

$$\mathbb{P}(N_1(h) \geq 2) \leq \mathbb{P}(N(h) \geq 2) = o(h).$$

Thus N_1 is a Poisson process with rate λp , and similarly N_2 has rate $\lambda(1 - p)$.

To see independence at a fixed time t , condition on $N(t) = m + n$:

$$\begin{aligned} \mathbb{P}(N_1(t) = m, N_2(t) = n) &= \binom{m+n}{m} p^m (1-p)^n e^{-\lambda t} \frac{(\lambda t)^{m+n}}{(m+n)!} \\ &= \left[e^{-\lambda p t} \frac{(\lambda p t)^m}{m!} \right] \left[e^{-\lambda(1-p)t} \frac{(\lambda(1-p)t)^n}{n!} \right]. \end{aligned}$$

The same factorization applied to increments proves independence of the two processes. $\square$

Example 5.4.2 (Immigration classification). Immigrants arrive in an area according to a Poisson process with rate 10 per week. Suppose each immigrant is of English descent with probability $1/12$, independently of all other arrivals. During four weeks, the number of immigrants of English descent is

$$\text{Poisson}\left(4 \cdot 10 \cdot \frac{1}{12}\right) = \text{Poisson}\left(\frac{10}{3}\right).$$

Hence the probability that there is no such arrival during the month is

$$\boxed{e^{-10/3}}.$$

5.4.2 Superposition

The splitting theorem may be read in reverse.

Proposition 5.4.3 (Superposition of homogeneous Poisson processes). *Let N_1 and N_2 be independent Poisson processes with rates λ_1 and λ_2 . Then*

$$N(t) := N_1(t) + N_2(t)$$

is a Poisson process with rate $\lambda_1 + \lambda_2$. Conditional on an event of the combined process, its label is type 1 with probability

$$p = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

and type 2 with probability $1 - p$, independently from event to event.

Proof. The sum has independent and stationary increments, and for each t ,

$$N(t) = N_1(t) + N_2(t) \sim \text{Poisson}((\lambda_1 + \lambda_2)t),$$

because independent Poisson variables add. The label probability follows from the splitting theorem, or directly from

$$\frac{\lambda_1 h + o(h)}{(\lambda_1 + \lambda_2)h + o(h)} \rightarrow \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

□

5.5 Applications of Splitting and Exponential Waiting Times

5.5.1 Poissonization of the coupon collector problem

There are m coupon types. Each coupon is type j with probability p_j , independently of previous coupons, where

$$p_j > 0, \quad \sum_{j=1}^m p_j = 1.$$

Let K be the number of coupons required to obtain all m types.

Introduce an independent rate-1 Poisson clock. At each event time, attach a coupon whose type has distribution $(p_1, \dots, p_m)$. Let $M_j(t)$ count type- j coupons by time t . By Poisson splitting, the processes $M_1, \dots, M_m$ are independent Poisson processes with rates $p_1, \dots, p_m$.

Let

$$X_j = \inf\{t : M_j(t) \geq 1\}$$

be the first arrival time of type j . Then

$$X_j \sim \text{Exp}(p_j), \quad X_1, \dots, X_m \text{ are independent.}$$

The time at which the collection becomes complete is

$$X = \max_{1 \leq j \leq m} X_j.$$

Therefore

$$\mathbb{P}(X \leq t) = \prod_{j=1}^m \mathbb{P}(X_j \leq t) = \prod_{j=1}^m (1 - e^{-p_j t}),$$

and the tail-integral formula gives

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t) dt = \int_0^\infty \left[1 - \prod_{j=1}^m (1 - e^{-p_j t}) \right] dt.$$

Let $T_1, T_2, \dots$ be the interarrival times of the rate-1 ground process. The coupon types are independent of these interarrival times, so K is independent of $(T_i)_{i \geq 1}$, and

$$X = \sum_{i=1}^K T_i.$$

Since $\mathbb{E}T_1 = 1$, the random-sum identity yields

$$\mathbb{E}X = (\mathbb{E}K)(\mathbb{E}T_1) = \mathbb{E}K.$$

Thus

$$\boxed{\mathbb{E}K = \int_0^\infty \left[1 - \prod_{j=1}^m (1 - e^{-p_j t}) \right] dt.}$$

5.5.2 Types appearing exactly once

Let I_i be the indicator that type i appears exactly once when the complete collection is first obtained. Then

$$\mathbb{E} \left[\sum_{i=1}^m I_i \right] = \sum_{i=1}^m \mathbb{P}(I_i = 1).$$

Let $S_i^{(2)}$ be the arrival time of the second type- i coupon. Since M_i has rate p_i ,

$$S_i^{(2)} \sim \text{Gamma}(2, p_i), \quad f_{S_i^{(2)}}(x) = p_i^2 x e^{-p_i x}, \quad x > 0.$$

The event $I_i = 1$ occurs exactly when every other type has appeared before the second type- i coupon arrives:

$$\{I_i = 1\} = \bigcap_{j \neq i} \{X_j < S_i^{(2)}\}.$$

Using independence of the split processes,

$$\begin{aligned}\mathbb{P}(I_i = 1) &= \int_0^\infty \mathbb{P}(X_j < x \text{ for all } j \neq i) f_{S_i^{(2)}}(x) dx \\ &= \int_0^\infty \left[\prod_{j \neq i} (1 - e^{-p_j x}) \right] p_i^2 x e^{-p_i x} dx.\end{aligned}$$

Hence

$$\mathbb{E} \left[\sum_{i=1}^m I_i \right] = \sum_{i=1}^m \int_0^\infty \left[\prod_{j \neq i} (1 - e^{-p_j x}) \right] p_i^2 x e^{-p_i x} dx.$$

5.5.3 Races between independent exponential clocks

Proposition 5.5.1 (Two exponential clocks). *If*

$$X_1 \sim \text{Exp}(\lambda_1), \quad X_2 \sim \text{Exp}(\lambda_2), \quad X_1 \perp\!\!\!\perp X_2,$$

then

$$\mathbb{P}(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Proof. Conditioning on $X_1 = x$,

$$\begin{aligned}\mathbb{P}(X_1 < X_2) &= \int_0^\infty \mathbb{P}(X_2 > x) \lambda_1 e^{-\lambda_1 x} dx \\ &= \int_0^\infty e^{-\lambda_2 x} \lambda_1 e^{-\lambda_1 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}.\end{aligned}$$

□

Proposition 5.5.2 (Minimum and winning clock). *Let $X_1, \dots, X_n$ be independent, with*

$$X_i \sim \text{Exp}(\lambda_i).$$

Then

$$X_{\min} := \min_{1 \leq i \leq n} X_i \sim \text{Exp} \left(\sum_{i=1}^n \lambda_i \right),$$

and

$$\mathbb{P}(X_{\min} = X_i) = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j}.$$

Proof. For $x \geq 0$,

$$\mathbb{P}(X_{\min} > x) = \prod_{i=1}^n \mathbb{P}(X_i > x) = \exp \left(-x \sum_{i=1}^n \lambda_i \right).$$

This proves the minimum law. Moreover,

$$\min_{j \neq i} X_j \sim \text{Exp} \left(\sum_{j \neq i} \lambda_j \right)$$

and is independent of X_i . Applying the two-clock formula gives the stated winning probability. □

5.5.4 Which Poisson process reaches its target first?

Let N_1 and N_2 be independent Poisson processes with rates λ_1 and λ_2 . Write $S_n^{(1)}$ for the time of the n th event of process 1, and $S_m^{(2)}$ for the time of the m th event of process 2.

The superposed process has rate $\lambda_1 + \lambda_2$, and each combined event is of type 1 with probability

$$p = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Thus comparing the two arrival times becomes a Bernoulli counting problem.

For $n = m = 1$,

$$\mathbb{P}(S_1^{(1)} < S_1^{(2)}) = p.$$

For $n = 2$ and $m = 1$, the first two combined events must both be type 1, so

$$\mathbb{P}(S_2^{(1)} < S_1^{(2)}) = p^2.$$

In general, $S_n^{(1)} < S_m^{(2)}$ exactly when at least n of the first $n + m - 1$ combined events are type 1. Therefore

$$\mathbb{P}(S_n^{(1)} < S_m^{(2)}) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n+m-1-k}.$$

5.6 Conditional Arrival Times and Order Statistics

5.6.1 One arrival conditioned on the count

Suppose N is a Poisson process with rate λ and $0 \leq s \leq t$. Conditional on $N(t) = 1$, the unique arrival time is uniform on $[0, t]$.

Proposition 5.6.1 (One conditional arrival).

$$S_1 \mid \{N(t) = 1\} \sim \text{Unif}[0, t].$$

Proof. Using independent increments,

$$\begin{aligned} \mathbb{P}(S_1 \leq s \mid N(t) = 1) &= \frac{\mathbb{P}(N(s) = 1, N(t) - N(s) = 0)}{\mathbb{P}(N(t) = 1)} \\ &= \frac{\lambda s e^{-\lambda s} e^{-\lambda(t-s)}}{\lambda t e^{-\lambda t}} = \frac{s}{t}. \end{aligned}$$

This is the c.d.f. of $\text{Unif}[0, t]$. □

5.6.2 Order statistics

Definition 5.6.2 (Order statistics). Let $Y_1, \dots, Y_n$ be real-valued random variables. Their order statistics are

$$Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(n)},$$

where $Y_{(k)}$ is the k th smallest value among $Y_1, \dots, Y_n$.

Proposition 5.6.3 (Joint density of all order statistics). *Let $Y_1, \dots, Y_n$ be iid with density f . Then*

$$f_{Y_{(1)}, \dots, Y_{(n)}}(y_1, \dots, y_n) = n! \prod_{i=1}^n f(y_i)$$

for $y_1 < y_2 < \dots < y_n$, and the density is zero outside this ordered region.

Proof. For distinct $y_1 < \dots < y_n$, the ordered vector is near $(y_1, \dots, y_n)$ whenever the original vector $(Y_1, \dots, Y_n)$ is near one of the $n!$ permutations of these values. The iid joint density is the same for every permutation and equals $\prod_i f(y_i)$. Summing the $n!$ equal contributions gives the formula. $\square$

In particular, if $Y_1, \dots, Y_n$ are iid $\text{Unif}[0, t]$, then

$$f_{Y_{(1)}, \dots, Y_{(n)}}(y_1, \dots, y_n) = \frac{n!}{t^n}, \quad 0 < y_1 < \dots < y_n < t.$$

5.6.3 Arrival Times Conditioned on the Count

Theorem 5.6.4 (Conditional arrival times are uniform order statistics). *Given $N(t) = n$, the arrival-time vector*

$$(S_1, S_2, \dots, S_n)$$

has the same distribution as the order statistics of n independent $\text{Unif}[0, t]$ random variables. Equivalently,

$$f_{S_1, \dots, S_n | N(t)=n}(s_1, \dots, s_n) = \frac{n!}{t^n}$$

for

$$0 < s_1 < s_2 < \dots < s_n < t.$$

Proof. For $0 < s_1 < \dots < s_n < t$, write

$$T_1 = s_1, \quad T_2 = s_2 - s_1, \quad \dots, \quad T_n = s_n - s_{n-1}.$$

The event $N(t) = n$ also requires no arrival in $(s_n, t]$. The joint density contribution is therefore

$$\lambda e^{-\lambda s_1} \lambda e^{-\lambda(s_2 - s_1)} \dots \lambda e^{-\lambda(s_n - s_{n-1})} e^{-\lambda(t - s_n)} = \lambda^n e^{-\lambda t}.$$

Dividing by

$$\mathbb{P}(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

gives $n!/t^n$ on the ordered simplex, exactly the density of uniform order statistics. $\square$

Remark 5.6.5 (Ordered and unordered formulations). If $Y_1, \dots, Y_n$ are iid $\text{Unif}[0, t]$, then

$$(S_1, \dots, S_n) \mid \{N(t) = n\} \stackrel{d}{=} (Y_{(1)}, \dots, Y_{(n)}).$$

As unordered sets,

$$\{S_1, \dots, S_n\} \mid \{N(t) = n\} \stackrel{d}{=} \{Y_1, \dots, Y_n\}.$$

5.6.4 Earlier Arrivals Conditioned on the n th Arrival Time

Because $\mathbb{P}(S_n = t) = 0$, the following statement is understood through a regular conditional density.

Proposition 5.6.6 (Conditioning on $S_n = t$). *Given $S_n = t$, the vector*

$$(S_1, \dots, S_{n-1})$$

has the same distribution as the order statistics of $n - 1$ independent $\text{Unif}[0, t]$ random variables. Its conditional density is

$$f_{S_1, \dots, S_{n-1} | S_n = t}(s_1, \dots, s_{n-1}) = \frac{(n-1)!}{t^{n-1}}$$

for

$$0 < s_1 < \dots < s_{n-1} < t.$$

Proof. The joint density of $(S_1, \dots, S_n)$ is

$$f_{S_1, \dots, S_n}(s_1, \dots, s_n) = \lambda^n e^{-\lambda s_n}, \quad 0 < s_1 < \dots < s_n.$$

Since

$$f_{S_n}(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t},$$

the conditional-density ratio is

$$\frac{\lambda^n e^{-\lambda t}}{\lambda^n t^{n-1} e^{-\lambda t} / (n-1)!} = \frac{(n-1)!}{t^{n-1}}.$$

□

5.7 Nonhomogeneous Poisson Processes

5.7.1 Time-dependent intensity

In many applications, the event rate changes with time. Stationary increments are then inappropriate.

Definition 5.7.1 (Nonhomogeneous Poisson process). A counting process $\{N(t) : t \geq 0\}$ is a *nonhomogeneous Poisson process* with intensity function $\lambda(t) \geq 0$ if

1. $N(0) = 0$;
2. the process has independent increments;
3. for each fixed t , as $h \downarrow 0$,

$$\mathbb{P}(N(t+h) - N(t) = 1) = \lambda(t)h + o(h);$$

4. for each fixed t , as $h \downarrow 0$,

$$\mathbb{P}(N(t+h) - N(t) \geq 2) = o(h).$$

When $\lambda(t) \equiv \lambda$ is constant, this reduces to the homogeneous Poisson process.

5.7.2 Time-dependent thinning

Proposition 5.7.2 (Thinning with a time-dependent retention probability). *Let $\tilde{N}$ be a homogeneous Poisson process with rate Λ . Suppose an event occurring at time t is retained with probability $p(t)$, independently of all other events and marks. Then the retained process is a nonhomogeneous Poisson process with intensity*

$$\lambda(t) = \Lambda p(t).$$

Proof. During $(t, t + h]$, the ground process has exactly one event with probability $\Lambda h + o(h)$. That event is retained with probability $p(t) + o(1)$, so

$$\mathbb{P}(N(t + h) - N(t) = 1) = \Lambda p(t)h + o(h).$$

Two or more retained events require two or more ground events and therefore have probability $o(h)$. Independent increments are preserved because the ground-process increments and all marks are independent on disjoint intervals. $\square$

5.7.3 Superposition and identification of the source

Proposition 5.7.3 (Superposition of nonhomogeneous Poisson processes). *Let N and M be independent nonhomogeneous Poisson processes with intensities $\lambda(t)$ and $\mu(t)$. Then*

$$N^*(t) := N(t) + M(t)$$

is a nonhomogeneous Poisson process with intensity

$$\lambda(t) + \mu(t).$$

Conditional on an event of N^ occurring at time t , the event came from N with probability*

$$\boxed{\frac{\lambda(t)}{\lambda(t) + \mu(t)}}.$$

Proof. The probability of one N -event and no M -event in $(t, t + h]$ is

$$(\lambda(t)h + o(h))(1 - \mu(t)h + o(h)) = \lambda(t)h + o(h).$$

Similarly, one M -event contributes $\mu(t)h + o(h)$, while simultaneous or multiple events have probability $o(h)$. Hence the combined intensity is $\lambda(t) + \mu(t)$. Dividing the first contribution by the total one-event probability gives the source probability. $\square$

Corollary 5.7.4 (Dominating-process construction). *Suppose $\lambda(t) \leq \Lambda$ for all $t \geq 0$. Let M be an independent nonhomogeneous Poisson process with intensity $\Lambda - \lambda(t)$. Then*

$$N^*(t) = N(t) + M(t)$$

is a homogeneous Poisson process with rate Λ , and N can be obtained from N^ by retaining an event at time t with probability*

$$\frac{\lambda(t)}{\Lambda}.$$

This is the time-dependent analogue of ordinary Poisson thinning.

5.7.4 Time-dependent classification into several types

Proposition 5.7.5 (Several time-dependent types). *Let N be a homogeneous Poisson process with rate Λ . At an event time s , assign the event to type $i \in \{1, \dots, k\}$ with probability $p_i(s)$, where*

$$p_i(s) \geq 0, \quad \sum_{i=1}^k p_i(s) = 1.$$

Assume all classifications are conditionally independent given their event times. Let $N_i(t)$ count type- i events by time t . Then $N_1, \dots, N_k$ are independent nonhomogeneous Poisson processes with intensities

$$\lambda_i(s) = \Lambda p_i(s).$$

In particular,

$$N_i(t) \sim \text{Poisson}\left(\Lambda \int_0^t p_i(s) \, ds\right)$$

and

$$\mathbb{E}N_i(t) = \Lambda \int_0^t p_i(s) \, ds.$$

Proof. Conditional on $N(t) = n$, the n event times, viewed without order, are iid $\text{Unif}[0, t]$. Therefore an arbitrary event in $[0, t]$ is type i with probability

$$P_i(t) = \frac{1}{t} \int_0^t p_i(s) \, ds.$$

Conditional on $N(t) = n$, the type counts have a multinomial distribution with probabilities $P_1(t), \dots, P_k(t)$. Multiplying this multinomial probability by the $\text{Poisson}(\Lambda t)$ probability of the total count and simplifying gives

$$\mathbb{P}(N_1(t) = n_1, \dots, N_k(t) = n_k) = \prod_{i=1}^k \left[e^{-\Lambda t P_i(t)} \frac{(\Lambda t P_i(t))^{n_i}}{n_i!} \right].$$

Thus the counts are independent Poisson variables with means

$$\Lambda t P_i(t) = \Lambda \int_0^t p_i(s) \, ds.$$

Applying the same argument to disjoint intervals gives the process-level conclusion. □

5.7.5 The mean-value function

Definition 5.7.6 (Mean-value function). For a nonhomogeneous Poisson process with intensity $\lambda(t)$, define

$$m(t) = \int_0^t \lambda(s) \, ds.$$

The function m is called the *mean-value function* or cumulative intensity.

Theorem 5.7.7 (Distribution of nonhomogeneous increments). *For $0 \leq s < t$,*

$$N(t) - N(s) \sim \text{Poisson}(m(t) - m(s)) = \text{Poisson}\left(\int_s^t \lambda(u) \, du\right).$$

In particular,

$$\mathbb{E}N(t) = \text{Var}(N(t)) = m(t) = \int_0^t \lambda(s) \, ds.$$

Proof. On a bounded time interval, choose Λ dominating $\lambda(t)$ and obtain N by thinning a rate- Λ process with retention probability $p(t) = \lambda(t)/\Lambda$. The time-dependent classification result applied on $(s, t]$ shows that the retained count is Poisson with mean

$$\Lambda \int_s^t p(u) \, du = \int_s^t \lambda(u) \, du.$$

□

Example 5.7.8 (Customer arrivals at a hot-dog stand). Measure time in hours after 8:00 a.m. Suppose the customer-arrival intensity is

$$\lambda(t) = \begin{cases} 5 + 5t, & 0 \leq t \leq 3, \\ 20, & 3 < t \leq 5, \\ 20 - 2(t - 5), & 5 < t \leq 9. \end{cases}$$

The interval from 8:30 a.m. to 9:30 a.m. corresponds to $t \in [1/2, 3/2]$. The number of arrivals in that interval is Poisson with mean

$$\int_{1/2}^{3/2} (5 + 5t) \, dt = 10.$$

Therefore

$$N(3/2) - N(1/2) \sim \text{Poisson}(10),$$

and the probability of no arrival during that hour is

$$e^{-10}.$$

Chapter 5 Summary of Key Formulas

Exponential distribution and memorylessness

For $X \sim \text{Exp}(\lambda)$,

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0, \infty)}(x), \quad \mathbb{P}(X > x) = e^{-\lambda x},$$

$$M_X(u) = \frac{\lambda}{\lambda - u} \quad (u < \lambda), \quad \mathbb{E}X = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

The memoryless identity is

$$\mathbb{P}(X > s + t \mid X > t) = \mathbb{P}(X > s).$$

Equivalently, its short-time event probability is

$$\mathbb{P}(t < X \leq t + h \mid X > t) = \lambda h + o(h).$$

Homogeneous Poisson process

For a rate- λ Poisson process,

$$N(t) - N(s) \sim \text{Poisson}(\lambda(t - s)), \quad 0 \leq s < t,$$

and disjoint increments are independent. Thus

$$\mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad \mathbb{E}N(t) = \text{Var}(N(t)) = \lambda t,$$

$$\mathbb{E}[e^{uN(t)}] = \exp\{\lambda t(e^u - 1)\}.$$

For small h ,

$$\mathbb{P}(N(h) = 1) = \lambda h + o(h), \quad \mathbb{P}(N(h) \geq 2) = o(h).$$

Interarrival and arrival times

If T_n is the n th interarrival time and S_n is the n th arrival time, then

$$T_1, T_2, \dots \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda), \quad S_n = \sum_{j=1}^n T_j \sim \text{Gamma}(n, \lambda),$$

$$N(t) = \sum_{n=1}^{\infty} \mathbf{1}_{\{S_n \leq t\}}.$$

Moreover,

$$\mathbb{P}(S_n \leq t) = 1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!},$$

$$f_{S_n}(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t > 0.$$

Splitting and superposition

If each event of a rate- λ process is independently labeled type 1 with probability p , then the type processes are independent with rates

$$\lambda p \quad \text{and} \quad \lambda(1-p).$$

Conversely, if N_1, N_2 are independent with rates λ_1, λ_2 , then

$$N_1 + N_2 \text{ has rate } \lambda_1 + \lambda_2,$$

and a combined event is from process 1 with probability

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Independent exponential races

For independent $X_i \sim \text{Exp}(\lambda_i)$,

$$\min_i X_i \sim \text{Exp}\left(\sum_i \lambda_i\right), \quad \mathbb{P}\left(X_i = \min_j X_j\right) = \frac{\lambda_i}{\sum_j \lambda_j}.$$

For two clocks,

$$\mathbb{P}(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

If $S_n^{(1)}$ and $S_m^{(2)}$ are target arrival times of two independent Poisson processes, then with

$$p = \frac{\lambda_1}{\lambda_1 + \lambda_2},$$

$$\mathbb{P}(S_n^{(1)} < S_m^{(2)}) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} p^k (1-p)^{n+m-1-k}.$$

Poissonized coupon collector

For coupon probabilities $p_1, \dots, p_m$, let K be the number required to obtain every type. Independent splitting gives first-arrival times

$$X_j \sim \text{Exp}(p_j), \quad X = \max_j X_j.$$

Hence

$$\mathbb{E}K = \int_0^\infty \left[1 - \prod_{j=1}^m (1 - e^{-p_j t}) \right] dt.$$

If I_i indicates that type i appears exactly once at completion, then

$$\mathbb{P}(I_i = 1) = \int_0^\infty \left[\prod_{j \neq i} (1 - e^{-p_j x}) \right] p_i^2 x e^{-p_i x} dx.$$

Conditional arrival times and order statistics

Given $N(t) = n$,

$$(S_1, \dots, S_n) \stackrel{d}{=} (Y_{(1)}, \dots, Y_{(n)}),$$

where $Y_1, \dots, Y_n$ are iid $\text{Unif}[0, t]$. The conditional joint density is

$$f_{S_1, \dots, S_n | N(t)=n}(s_1, \dots, s_n) = \frac{n!}{t^n}, \quad 0 < s_1 < \dots < s_n < t.$$

Given $S_n = t$,

$$f_{S_1, \dots, S_{n-1} | S_n=t} = \frac{(n-1)!}{t^{n-1}}$$

on $0 < s_1 < \dots < s_{n-1} < t$.

Nonhomogeneous Poisson process

For intensity $\lambda(t)$, define

$$m(t) = \int_0^t \lambda(s) \, ds.$$

Then

$$N(t) - N(s) \sim \text{Poisson}(m(t) - m(s)),$$

$$\mathbb{E}N(t) = \text{Var}(N(t)) = m(t).$$

A rate- Λ homogeneous process thinned at time t with retention probability $p(t)$ has intensity

$$\lambda(t) = \Lambda p(t).$$

Independent nonhomogeneous processes with intensities $\lambda(t)$ and $\mu(t)$ superpose to intensity $\lambda(t) + \mu(t)$, and a combined event at time t came from the first process with probability

$$\frac{\lambda(t)}{\lambda(t) + \mu(t)}.$$